

UPSC Prelims

Environment & Ecology

Class No. 8

Climate Change

- **Climate change** refers to **long-term shifts** in temperatures and weather (rainfall, snow, and wind) patterns attributed **directly or indirectly to human** (anthropogenic) activity that alters the composition of the global atmosphere causing **natural climate variability** observed over comparable periods (hundreds of years).
- Climate change encompasses **global warming**, but refers to a broader range of changes (due to variability in temperature, rainfall, wind and water patterns), such as rising sea levels, shrinking mountain glaciers, accelerating loss of cryosphere (melting of ice caps of mountains and polar regions), ocean acidification, etc.

Climate Change



- Global warming refers to **long-term warming** (rise in global temperatures) of the planet, which is only one aspect of climate change.
- Throughout its evolutionary history, earth has experienced episodic climate change events involving global warming and global cooling (glaciation events).
- Global temperature shows a well-documented rise (global warming) since the early 20th century & most notably since the late 1970s.

Greenhouse Gas Effect



- A greenhouse is a structure whose roof & walls are made chiefly of transparent material, such as glass, in which plants requiring regulated climatic conditions are grown.
- In a greenhouse, the incident solar radiation (the visible and adjacent portions of the infrared and ultraviolet ranges of the spectrum) passes through the glass roof and walls. It is absorbed by the floor, earth, and contents, which become warmer and re-emit the energy as longer-wavelength infrared radiation.

Greenhouse Gas Effect



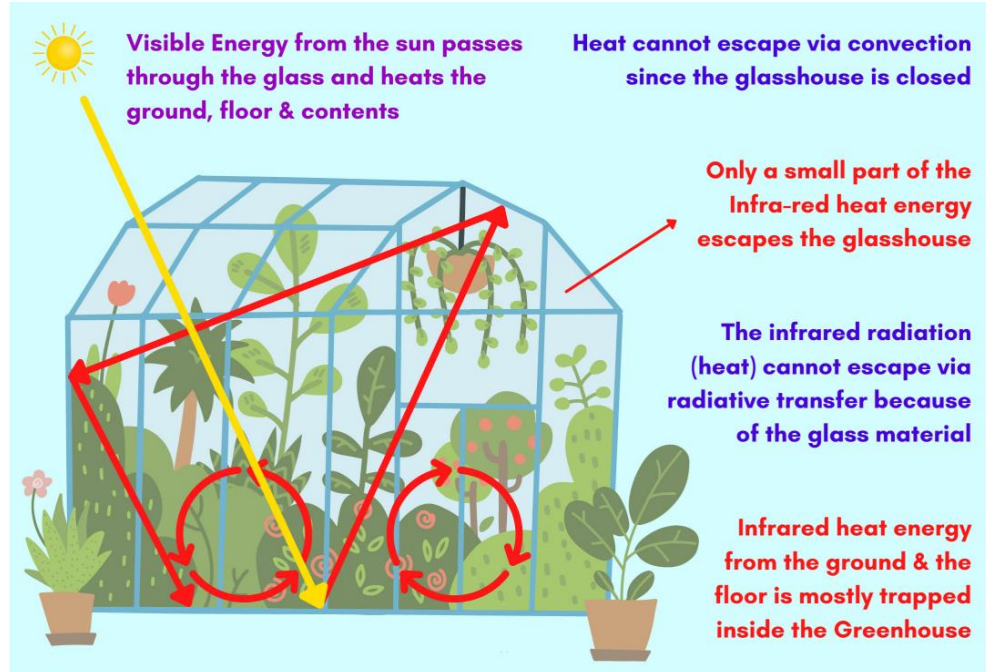
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Greenhouse Gas Effect



- Glass and other materials used for greenhouse walls do not transmit infrared radiation, so infrared radiation cannot escape via radiative transfer. As the structure is not open to the atmosphere, heat also cannot escape via convection, so the temperature inside the greenhouse rises. This is known as the greenhouse effect.

Greenhouse Gas Effect

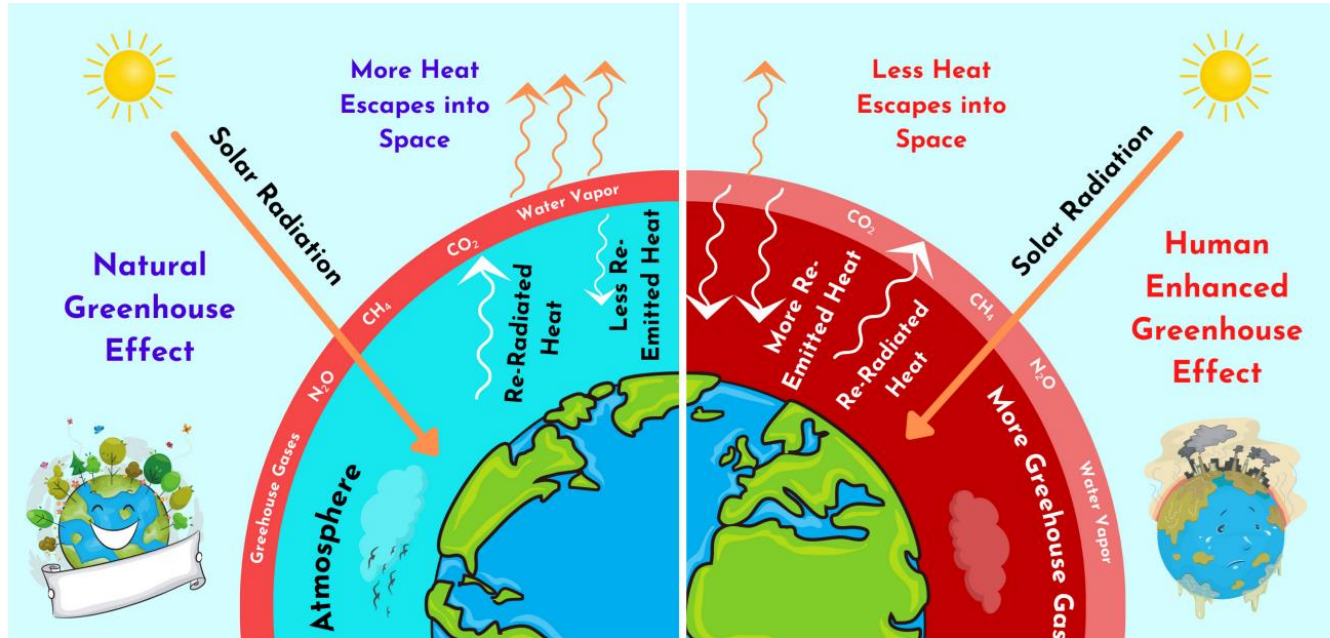


Greenhouse Gas Effect



- The greenhouse effect is a natural process that occurs in the Earth's atmosphere, which helps to maintain a stable temperature on the planet's surface.
- It is caused by certain gases in the atmosphere, known as greenhouse gases, which trap some of the sun's heat and prevent it from escaping back into space. This process keeps the Earth's temperature at a comfortable level for living things.
- 95 % of the greenhouse effect is caused by water vapour and small water particles in the atmosphere.

Greenhouse Gas Effect



Importance of Natural Greenhouse Effect



- The greenhouse effect is a natural phenomenon occurring for millions of years on the earth. Life on the earth has been possible because of this natural greenhouse effect which is due to water vapour and small particles of water present in the atmosphere. Together, these produce more than 95 per cent of total greenhouse warming.
- Average global temperatures are maintained at about 15°C due to the natural greenhouse effect. Without this phenomenon, average global temperatures might have been around -17°C , and at such low temperatures, life would not be able to exist.

Greenhouse Gases



- Atmospheric gases like Carbon Dioxide (CO_2), Methane (CH_4), Nitrous Oxide (N_2O), Water Vapour, and Chlorofluorocarbons (CFCs) can trap the out-going infrared radiation from the earth's surface.
- Hence these gases are known as greenhouse gases (GHGs), and the heating effect is known as the greenhouse effect.
- Worldwide, since 1880, average temperature has risen 1°C relative to the baseline.
- This increased temperature can lead to deleterious environmental changes and result in odd climatic changes like the elnino, increased melting of cryosphere, rise in sea level, coral bleaching etc.

- **Global warming potential**
- It is a measure of how much energy the emissions of 1 ton of a gas will absorb over a given period of time, relative to the emissions of 1 ton of carbon dioxide (CO₂).
- The larger the GWP, the more that a given gas warms the Earth compared to CO₂ over that time period.
- The time period usually used for GWPs is 100 years.
- GWPs provide a common unit of measure, which allows analysts to add up emissions estimates of different gases (e.g., to compile a national GHG inventory), and allows policymakers to compare emissions reduction opportunities across sectors and gases

Greenhouse Gases



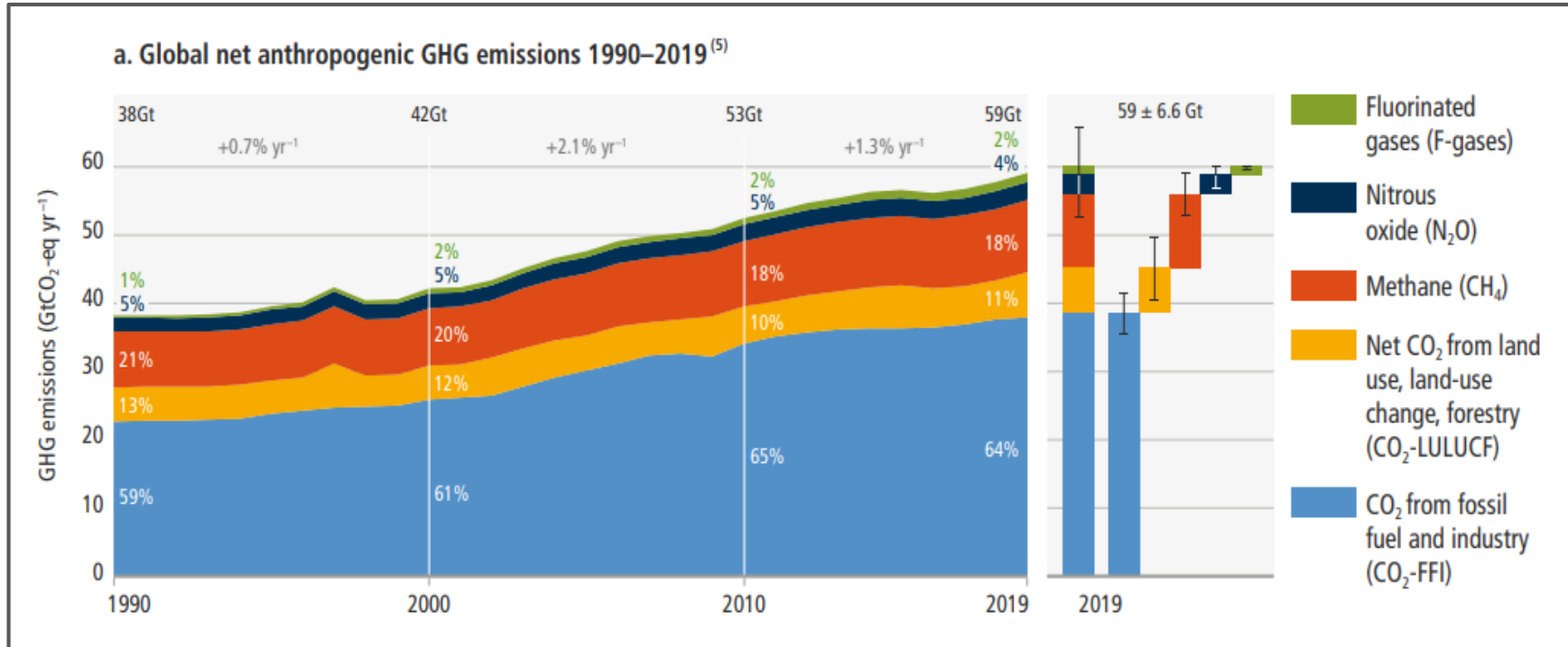
Gas	Global warming potential (100 years)	Lifetime (years)
Carbon dioxide (CO ₂)	1	50-200
Methane (CH ₄)	21	12
Nitrous oxide (N ₂ O)	310	120
Hydrofluorocarbons (HFCs)	140-11700	1-270
Perfluorocarbons (PFCs)	6500-9200	800-50000
Sulphur hexafluoride (SF ₆)	23900	3200

Greenhouse Gases



Gas	Sources
Carbon dioxide (CO ₂)	Burning of fossil fuels, deforestation, etc.
Carbon monoxide (CO)	Iron ore smelting, burning of fossil fuels, burning e-waste.
Methane (CH ₄)	Growing paddy, excreta of cattle and other livestock, termites, burning of fossil fuel, wood, landfills, wetlands, fertiliser factories, etc.
Nitrous oxide (N ₂ O)	Burning of fossil fuels, wood and crop residue; fertilisers.
Chlorofluorocarbons (CFCs)	Refrigeration, solvents, insulation foams, aero propellants, industrial and commercial uses.
Hydrofluorocarbons (HFCs)	Semiconductor manufacturing, other industries
Perfluorocarbons (PFCs)	Same as HFCs plus aluminium smelting
Sulphur hexafluoride (SF ₆)	Electrical transmission systems, magnesium and aluminium production

Greenhouse Gases



Source- IPCC AR6

- **Carbon Dioxide (CO₂)**
- It is a meteorologically important gas as it is transparent to incoming solar radiation but opaque to outgoing terrestrial radiation.
- Being an efficient absorber of infrared radiation (heat), it absorbs a part of terrestrial radiation and reflects some of it towards the earth's surface.
- It is mainly responsible for the greenhouse effect and heat energy budget.
- Its concentration is greater close to the earth's surface as it is denser than air.
- Atmospheric carbon dioxide measured at NOAA's Mauna Loa Atmospheric Baseline Observatory (Hawaii) peaked for 2021 in May at a monthly average of 419 ppm.

Greenhouse Gases



- **Total carbon estimation**
- 1.85 Billion Gigatons (Gt) of total carbon on Earth

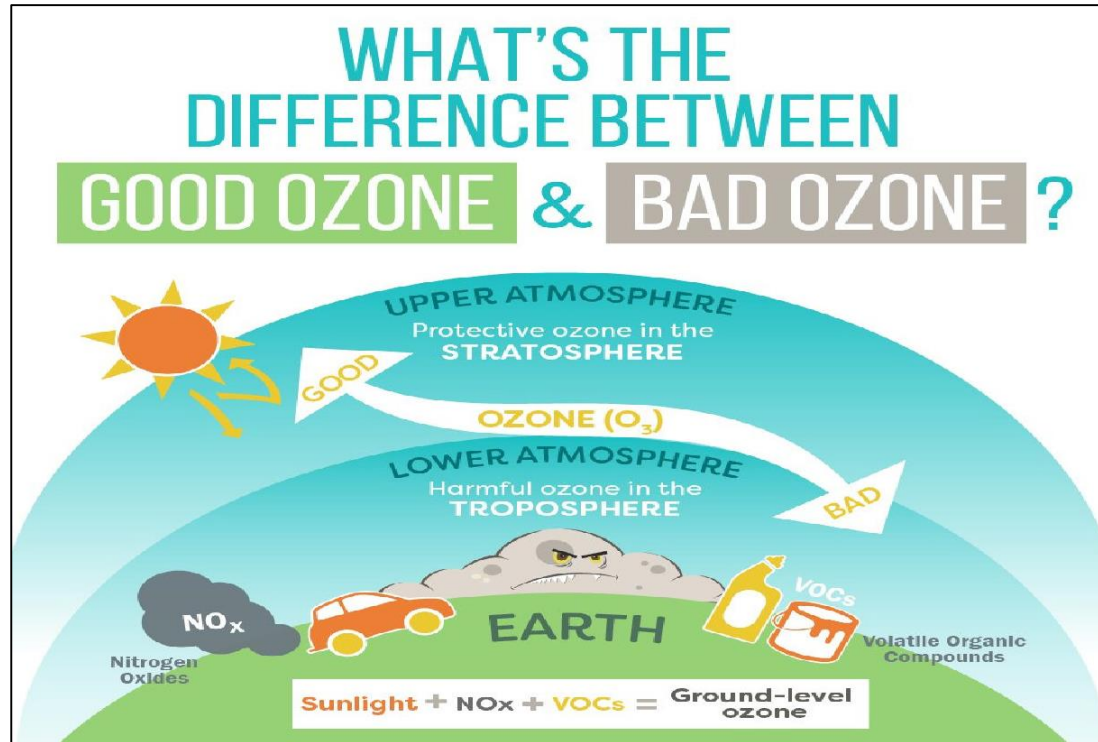
Below the surface	Above the surface
1.845 Billion Gt of the total carbon on Earth is below the surface.	43,500 Gt of the total carbon on Earth is above the surface.
Of this, 315 Million Gt of carbon is in continental and oceanic lithospheres.	1. 37,000 Gt in the deep ocean (85.1 %)
	2. 3,000 Gt in marine sediments (6.9%)
	3. 2,000 Gt in terrestrial biosphere (4.6%)
	4. 900 GT in the surface ocean (2%)
	5. 590 Gt in the atmosphere (1.4%)

- **Ozone**
- Ozone is another important greenhouse gas.
- Ozone (O_3) forms less than 0.00005% by volume of the atmosphere and is unevenly distributed.
- It is formed naturally in the stratosphere (good ozone — formed due to the interaction between O_2 and UV light) and absorbs harmful ultraviolet rays of the sun.
- However, at the ground level (bad ozone), it is a pollutant (short-lived greenhouse gas) with highly toxic effects.
- At ground level, pollutants like Carbon Monoxide (CO), Nitrogen Dioxide (NO_2) and Volatile Organic Compounds (VOC) convert O_2 to tropospheric ozone (O_3) in the presence of sunlight.

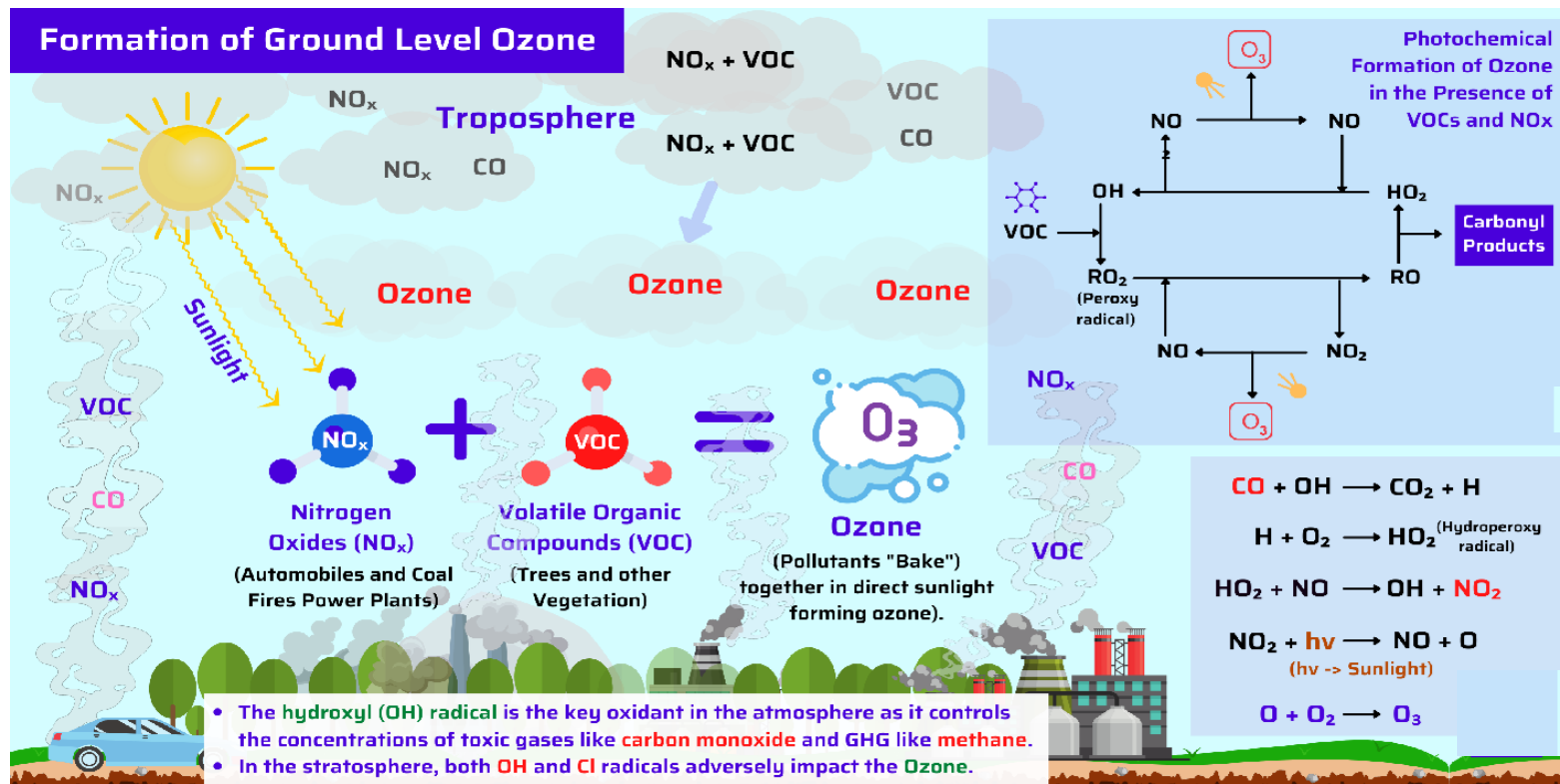
Greenhouse Gases



- **Ozone**
- Ground level ozone is not released directly. Some of it is transported from the stratosphere. Rest is formed by a chemical reaction
- At ground level, pollutants like Carbon Monoxide (CO), Nitrogen Dioxide (NO₂) and Volatile Organic Compounds (VOC) convert O₂ to tropospheric ozone (O₃) in the presence of sunlight.



Ground Level Ozone.



- **Tropospheric Ozone**
- Tropospheric ozone formation reactions
 1. Carbon Monoxide reacts with the hydroxyl radical (-OH) to produce hydroperoxy radical (HO_2).
 2. Volatile Organic Compounds react with hydroxyl radical (-OH) to produce peroxy radical (RO_2).
 3. Hydroperoxy radicals react with Nitrogen Oxide (NO) to form NO_2 & hydroxyl radical (OH) / alkoxy radical (RO).
 4. NO_2 gives O_3 via photolysis (molecule separation by light).

- **Harmful Effects of Ozone**
- Ozone at ground level is a harmful air pollutant and one of the main ingredients of smog.
- Ozone makes our eyes itchy and watery. It lowers our resistance to colds and pneumonia.
- Asthma patients are at the most significant risk from breathing air containing ozone.
- Elevated ozone exposure can affect sensitive vegetation and ecosystems, including forests, especially during the growing season.
- Ozone is most likely to reach unhealthy levels on **hot sunny days**. It can also be transported long distances by wind, so even rural areas can experience high O₃ levels.

- **Photochemical Smog**
- Photochemical smog is also known as summer smog or Los Angeles smog.
- It occurs most prominently in urban areas with high vehicular emissions.
- It forms when nitrogen oxides & volatile organic compounds react together in the presence of sunlight to form ozone (a secondary pollutant).
- The resulting smog causes a light brownish colouration of the atmosphere, reduced visibility, plant damage, irritation of the eyes, and respiratory distress.
- The atmospheric pollution level of Los Angeles, Beijing, Delhi, etc. is increased by inversion that traps smog near the ground.
- It is highly toxic to humans and can cause severe sickness and shorten life.
- Temperature inversions are accentuated, and precipitation is reduced due to smog.

Greenhouse Gases



$NO + VOC \text{ (volatile organic compounds)} \longrightarrow NO_2 \text{ (Nitrogen Dioxide)}$

$NO_2 + UV \text{ (sunlight)} \longrightarrow NO + O \text{ (Nitrogen Oxide + atomic oxygen)}$

$O + O_2 \longrightarrow O_3 \text{ (Ozone)}$

$NO_2 + VOC \longrightarrow PAN \text{ (peroxyacetyl nitrate - } C_2H_3NO_5)$

Result: $NO + VOC + O_2 + UV \longrightarrow O_3, PAN, \text{ and other oxidants like OH}$

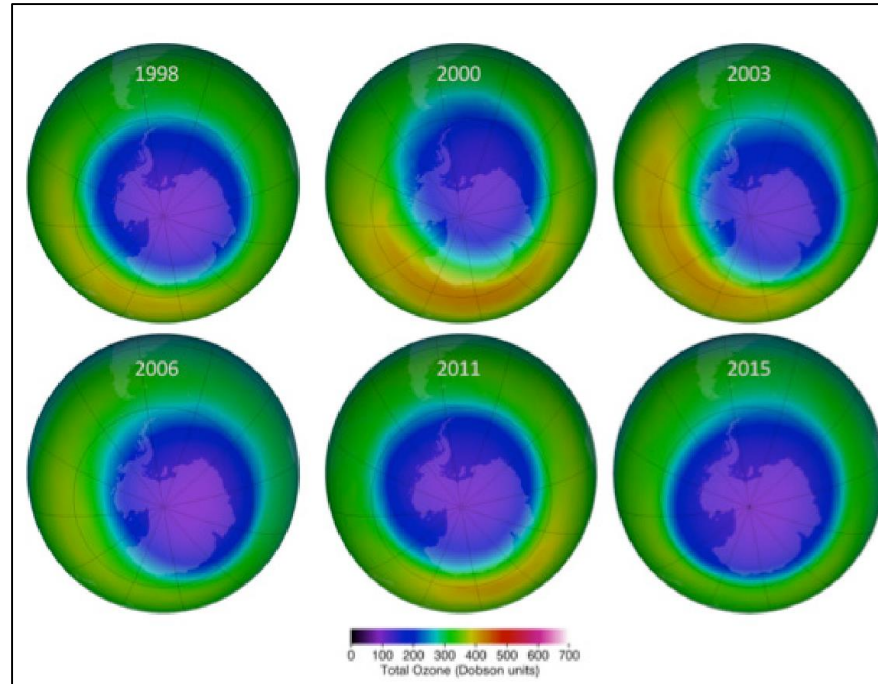
Chemical reaction involved in photochemical smog

- **Ozone Depletion**
- There is a steady decline in the total ozone volume in Earth's stratosphere. A much more significant decrease in stratospheric ozone is observed around Earth's polar regions.
- The unique cocktail of the powerful polar vortex and low temperatures generates stratospheric clouds that react with CFCs and other ozone-depleting substances and destroy the Ozone layer in the process.
- Remember : Chloride ion causes depletion of ozone layer in the stratosphere.

- **Stratospheric Ozone Depleting Substances (ODS)**
- ODS are human-made gases that release chlorine and bromine atoms on exposure to UV rays and destroy stratospheric ozone (good ozone). They include:
 1. chlorofluorocarbons (CFCs)
 2. hydrochlorofluorocarbons (HCFCs)
 3. hydrobromofluorocarbons (HBFCs)
 4. halons (halocarbon gases were used in fire extinguishers)

5. methyl bromide (was used as a fumigant for pest control — suffocating pests with poison)
 6. carbon tetrachloride (formerly widely used in fire extinguishers, refrigerants & as cleaning agents)
 7. methyl chloroform (aerosols, solvent for organic compounds; used for cleaning metals & circuit boards).
- ODS have been used as refrigerants in air conditioners and refrigerators, foam-blowing agents, components in electrical equipment, industrial solvents, solvents for dry cleaning, aerosol spray propellants and fumigants.

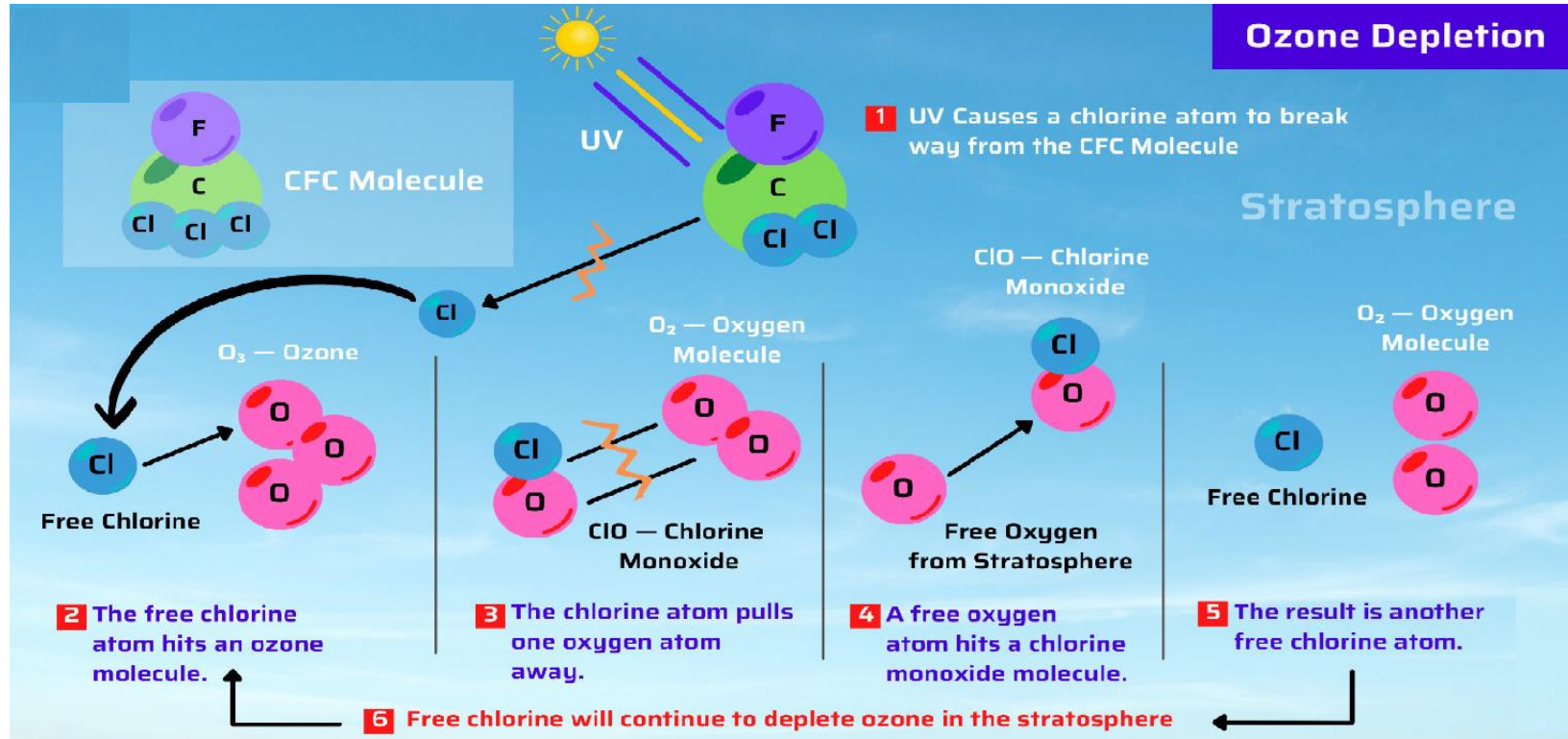
Greenhouse Gases



Ozone hole at South Pole

- **Process**
- Depletion of ozone is due to increased halocarbons in the atmosphere.
- Halocarbons are compounds in which the hydrogen of a hydrocarbon is replaced by halogens — a group of reactive non-metallic elements like fluorine, chlorine, bromine, iodine, etc.
- Halogen Atoms Like Chlorine Destroy Ozone
- Photodissociation (under the influence of sunlight) of ozone-depleting substances (CFCs, HCFCs, carbon tetrachloride and trichloroethane, freons, halons) like halocarbon refrigerants, solvents, propellants, and foam-blowing agents creates free chlorine atoms that destroy ozone.

Greenhouse Gases



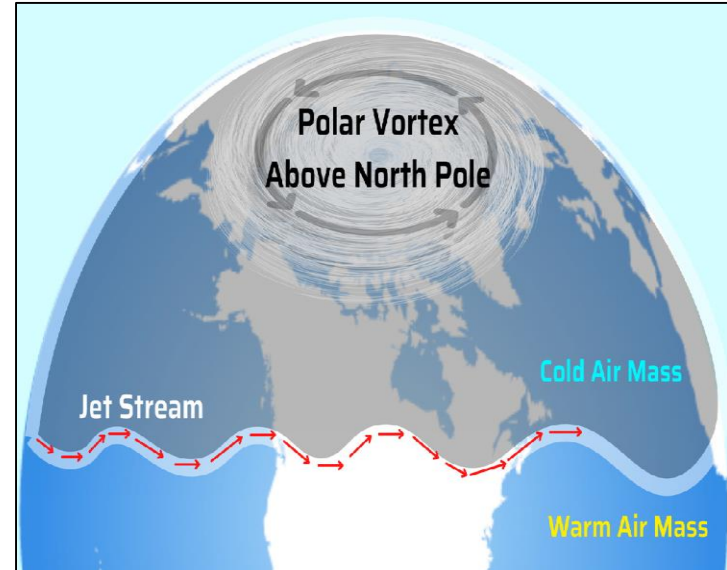
Greenhouse Gases



- PSCs or nacreous clouds contain water, nitric acid and sulfuric acid. Cl-catalysed ozone depletion is enhanced in the presence of polar stratospheric clouds.
- PSCs convert reservoir compounds into reactive free radicals (Cl and ClO), thereby significantly increasing the reactive halogen radicals.
- These free radicals accelerate the depletion of ozone.
- Polar Stratospheric Clouds (PSCs) are nacreous clouds that extend from 12-22 km above the surface. They are formed mainly during the polar vortex event in winter, more intense at the south pole.

Greenhouse Gases

- **Role of Polar Vortex and Polar Stratospheric clouds**
- Polar vortex (circumpolar vortex) is a polar cyclone. It gives rise to Polar Stratospheric Clouds (PSCs).
- Polar Vortex Acts as a Transporter for Halogens
- Polar Stratospheric Clouds (PSCs) are nacreous clouds that extend from 12-22 km above the surface.
- They are formed mainly during the polar vortex event in winter, more intense at the south pole.
- Nacreous clouds are rare clouds in the frigid regions of the lower stratosphere.

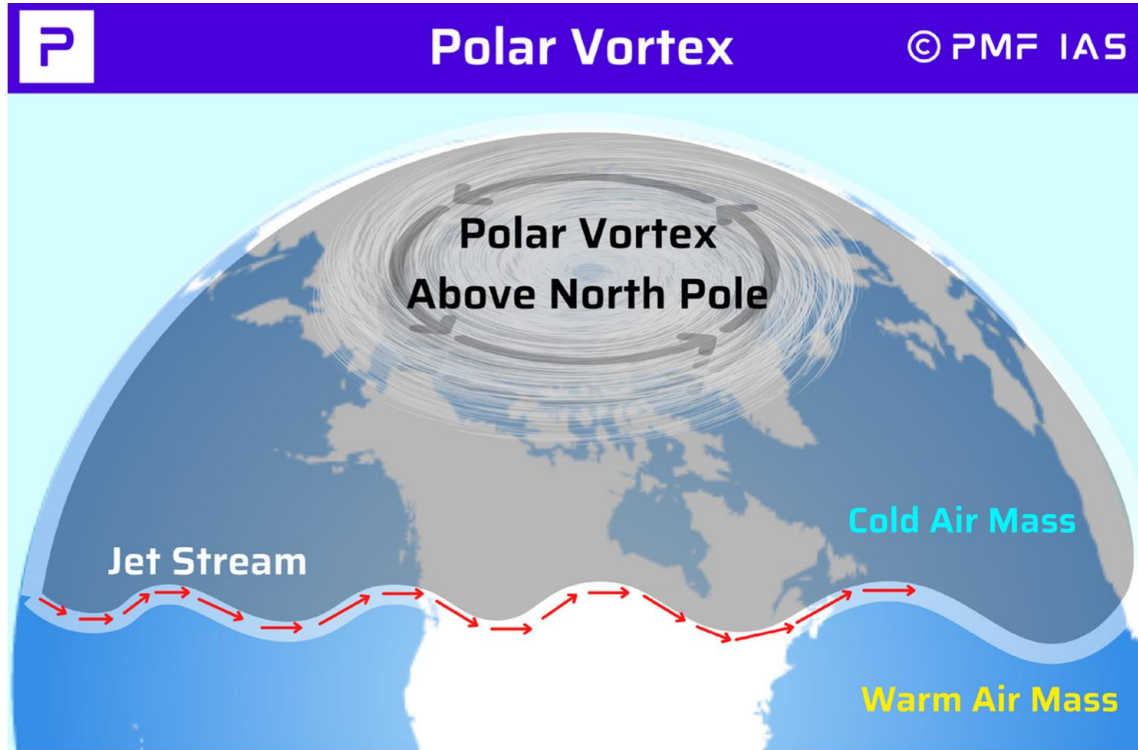


Greenhouse Gases

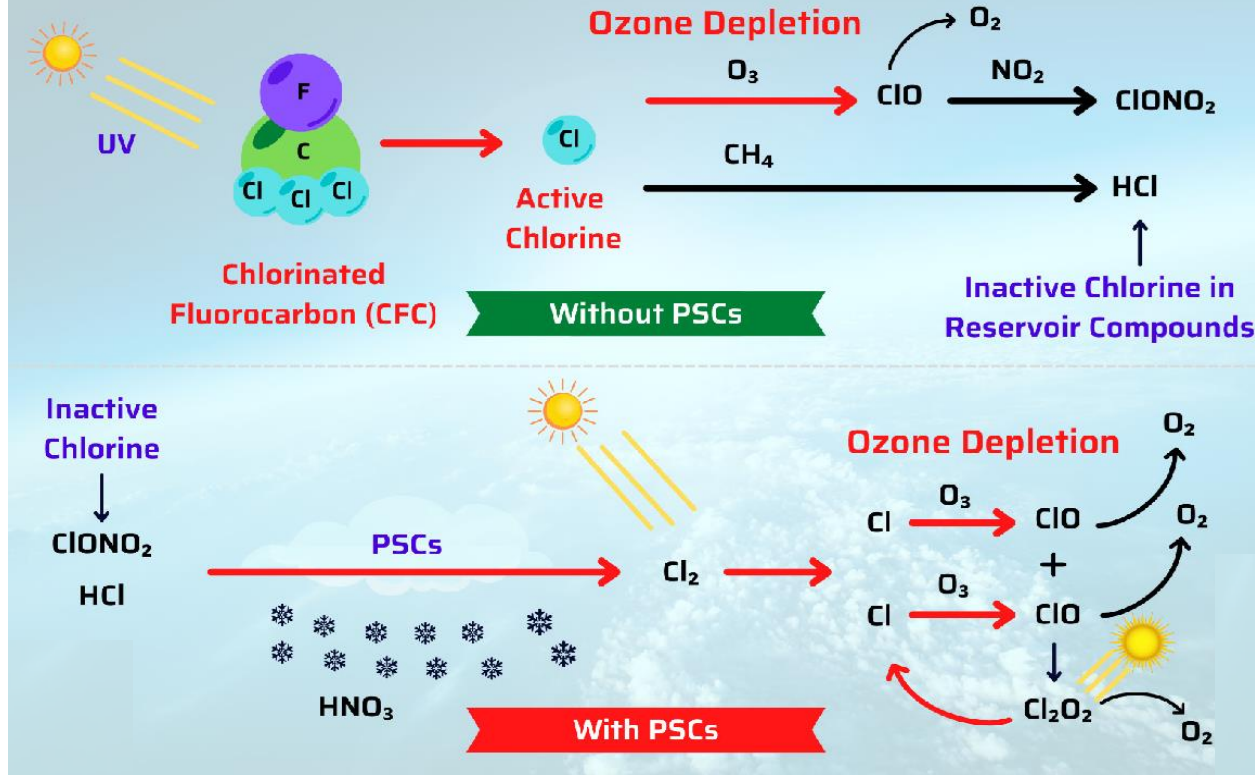


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Greenhouse Gases



Polar Stratospheric Clouds (PSCs) & Ozone Depletion



- **Impact of Nanoparticles (NPs)**
- The hydroxyl radical (OH) is the most important oxidant in the troposphere and lower stratosphere.
- OH initiates the removal of stratospheric ozone-depleting compounds such as hydrochlorofluorocarbons (HCFCs).
- NPs being very reactive, bind with OH and ultimately result in its reduction of OH radicals, maintaining the concentration of ozone-depleting substances.

Greenhouse Gases



- **Ozone hole over Antarctic vs Arctic:**
- Ozone holes are most commonly found over the Antarctic region (experiences strong polar vortex).
- They are comparatively rare & small over the Arctic (because the polar vortex here is comparatively weak and temperatures are relatively high).

Tropical Ozone Hole :

- Recently, a large, year-round ozone hole was discovered in the lower stratosphere over the tropics. It has been in the tropics since the 1980s but was discovered recently.
- The tropical Ozone Hole is an all-season ozone hole compared to the Antarctic Ozone Hole, which is pre- dominantly visible during springtime. It is seven times greater in area than the Antarctic ozone hole, and the depth of both is **almost the same**.

- **Short-lived climate pollutants**
- They are powerful climate forcers that remain in the atmosphere for a much shorter period of time than carbon dioxide (CO₂), yet their potential to warm the atmosphere can be many times greater.
- Certain short-lived climate pollutants are also dangerous air pollutants that have harmful effects for people, ecosystems and agricultural productivity.
- The short-lived climate pollutants- black carbon, methane, tropospheric ozone, and hydrofluorocarbons are the most important contributors to the man-made global greenhouse effect after carbon dioxide, responsible for up to 45% of current global warming.
- Black carbon is the second-largest contributor to warming the planet behind carbon dioxide (CO₂).

Greenhouse Gases

SHORT-LIVED CLIMATE POLLUTANTS

Response to mitigation efforts

SUBSTANCE	 ANTHROPOGENIC SOURCES	LIFETIME IN ATMOSPHERE	LOCAL REGIONAL GLOBAL		
					
IMPACTS/MITIGATION					
BLACK CARBON (BC)	    	DAYS			
METHANE (CH ₄)	    	12 YEARS			
TROPOSPHERIC OZONE (O ₃)	     	WEEKS			
HYDROFLUOROCARBONS (HFCs)	     	15 YEARS (WEIGHTED BY USAGE)			

www.ccacoalition.org

Greenhouse Gases



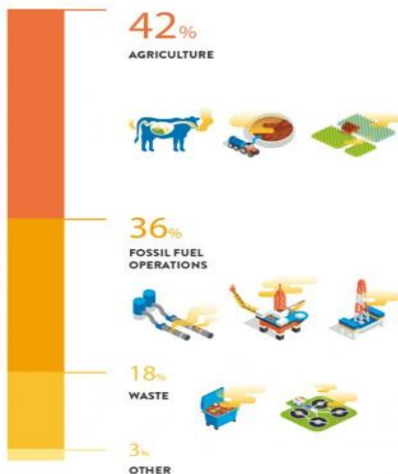
- **Methane (CH₄)**
- Characteristics:
- Colourless and odourless gas.
- Primary component of natural gas.
- Powerful GHG: Lifespan in atmosphere: 20 years (fewer than CO₂).
- Global Warming Potential: more than 80 times greater than that of CO₂.
- Responsible for:
- More than 25% of global warming.
- ~1/2 of the growth in tropospheric ozone formation.
- India among the top 5 methane emitters in world.

Greenhouse Gases

METHANE (CH_4)

SOURCES

Methane is one of the fastest growing greenhouse gases in the atmosphere. Human activity causes 2/3 of emissions.



% = global emissions

Methane emissions caused by human activities are one of the most significant drivers of climate change. Methane is also the main precursor of tropospheric ozone, a powerful greenhouse gas and air pollutant.

IMPACTS

CLIMATE

Responsible for 40% of warming since the industrial revolution

86x

times more powerful than carbon dioxide over a 20-year period

HEALTH

Increasing emissions are driving a rise in tropospheric ozone air pollution, which causes 1+ million premature deaths annually. Methane is responsible for roughly 1/2 of these deaths.

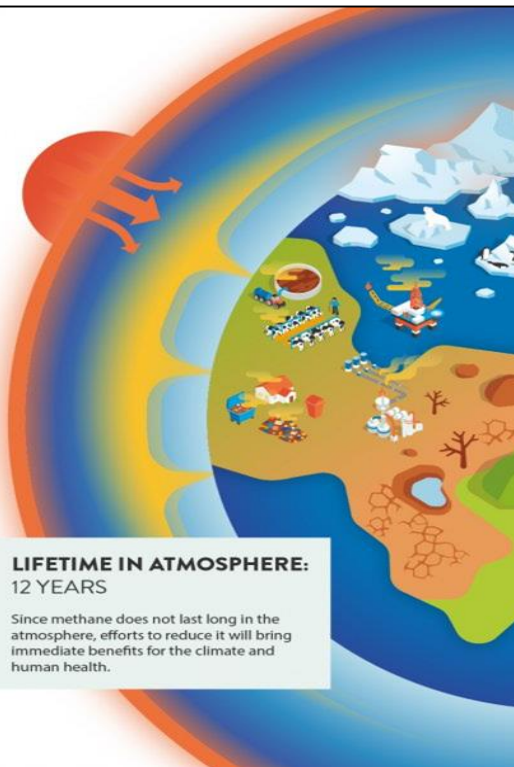


- Respiratory diseases
- Heart disease
- Damaged airways and lung tissue

AGRICULTURE & ECOSYSTEMS



Up to 15% annual yield losses of soy, wheat, rice and maize

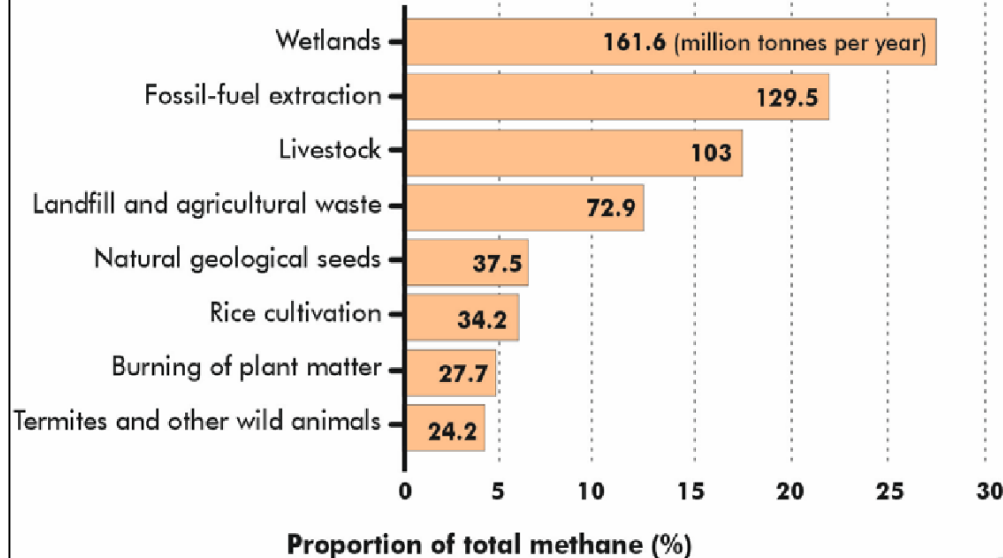


LIFETIME IN ATMOSPHERE: 12 YEARS

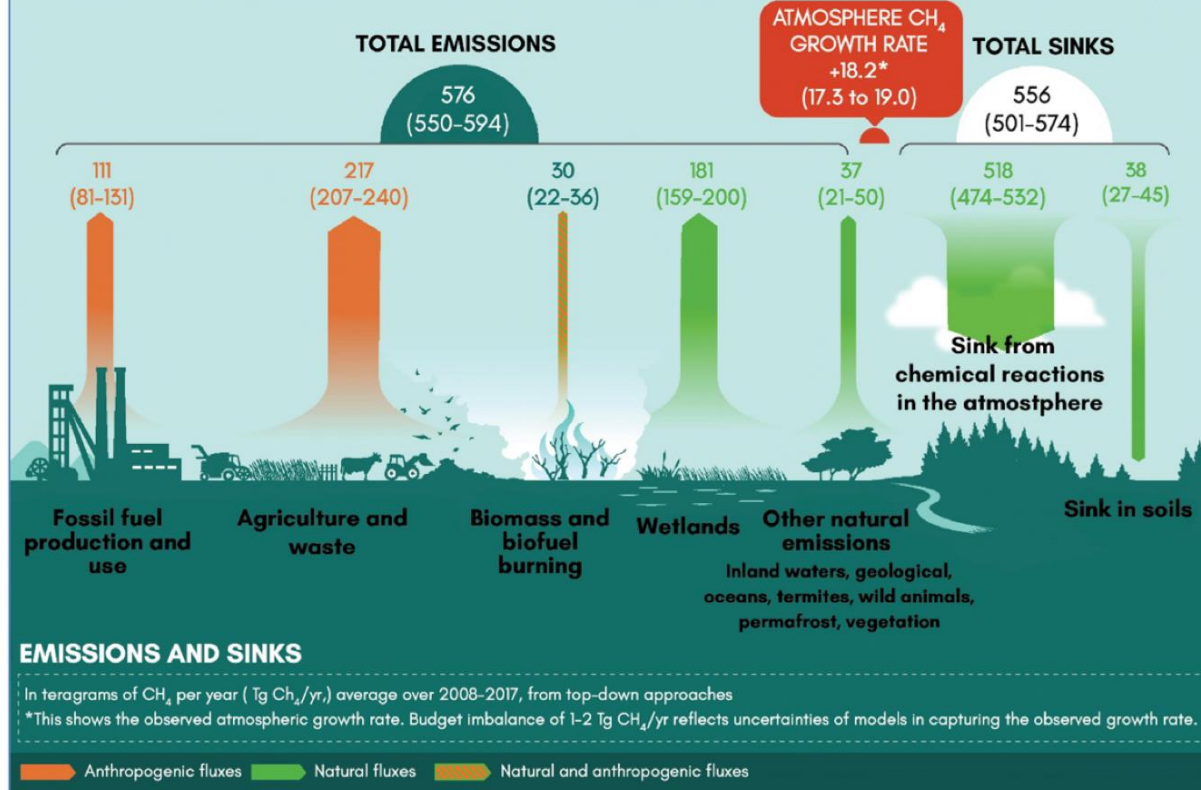
Since methane does not last long in the atmosphere, efforts to reduce it will bring immediate benefits for the climate and human health.

WHERE IS METHANE COMING FROM?

Most of the increase in emissions seems to be biological in origin, rather than having been released from below Earth's surface during the extraction of fossil fuels.



GLOBAL METHANE BUDGET 2008-2017



- **‘Global Methane Assessment: 2030 Baseline Report’**
- Released by Climate and Clean Air Coalition (CCAC) and United Nations Environment Programme (UNEP), launched at Climate and Clean Air ministerial meeting at COP 27 to the UNFCCC.
- Examines baseline projections of methane emissions over the coming decade and implications for Global Methane Pledge (GMP) target.
- Key findings:
- Amount of Methane in the atmosphere is 260% of pre-industrial levels.
- Human-driven methane emissions are responsible for nearly 45% of current net warming.
- 2021 saw the largest annual increase recorded.

Greenhouse Gases



- Earlier report titled “Global Methane Assessment: Benefits and Costs of Mitigating Methane Emissions” concluded that reduction of anthropogenic methane emissions by 45% would prevent a rise in global warming by up to 0.3 degrees Celsius by 2045. Also released by Climate and Clean Air Coalition along with UNEP.
- Global methane budget : By global carbon project.
- Agriculture and waste > Wetlands > Fossil Fuel.
- Flooded rice cultivation has been identified as one of the leading global agricultural sources of anthropogenic methane (CH₄) emissions



CLIMATE &
CLEAN AIR
COALITION
TO REDUCE SHORT-LIVED
CLIMATE POLLUTANTS

Climate and Clean Air Coalition



Genesis



- In 2012, the governments of **Bangladesh, Canada, Ghana, Mexico, Sweden and the United States**, along with the **United Nations Environment Programme (UNEP)**, came together to initiate efforts to treat **short-lived climate pollutants** as an urgent and collective challenge and formed the Climate & Clean Air Coalition.
 - The Coalition's **Secretariat** is hosted by the United Nations Environment Programme (UNEP).

Objective



- To improve air quality and protecting the climate through actions to reduce short-lived climate pollutants

Membership



- A voluntary partnership of governments, intergovernmental organizations, businesses, scientific institutions and civil society organizations.
 - **India is a State partner.**

Is India a member



- **Global Methane Pledge (GMP)**
- Launched at COP26 of UNFCCC in 2021 to catalyse action to reduce methane emissions.
- Led by: United States and European Union.
- Aim: To collectively reduce methane emissions by at least 30% below 2020 levels by 2030.
- Membership: More than 100 country participants, together responsible for 45% of global human-caused methane emissions.
- India did not sign the pledge because of its concerns over the impact on trade, on the country's vast farm sector, and the role of livestock in the rural economy.

- **International Methane Emissions Observatory (IMEO)**
- According to a report by IMEO, cutting methane emissions is fastest way to tackle climate change in short-term.
- It is An initiative by UNEP, launched in 2021, with support from European Commission
- Aim: To catalyse reduction of methane emissions, starting with fossil fuel sector.
- Core implementing partner of Global Methane Pledge

- **Global Energy Monitor (GEM)**
- As per GEM, just 30 fossil fuel companies account for nearly half of the planet-warming methane emitted by the world's energy sector.
- A non-profit organization which aims to develop and share information in support of the worldwide movement for clean energy.

Tropospheric Ozone

TROPOSPHERIC OZONE (O_3)

Tropospheric ozone is a powerful greenhouse gas and air pollutant that is harmful to human health, agricultural crops and ecosystems.

SOURCES Tropospheric ozone does not have any direct emissions sources, rather it is formed when sunlight interacts with different pollutants.

LIFETIME IN ATMOSPHERE: WEEKS

Reducing the pollutants that form tropospheric ozone would generate rapid benefits for the climate and human health.



Black Carbon (Soot)



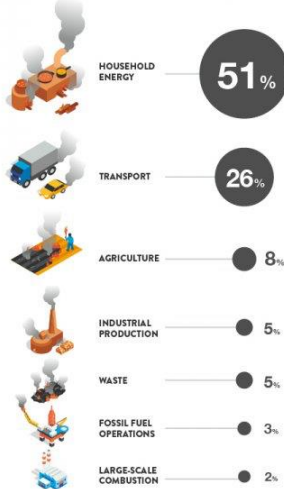
- Black carbon warms the earth by reducing albedo (the ability to reflect sunlight) when deposited on snow.
- It is the strongest absorber of sunlight (a lot more than carbon dioxide) and heats the air directly.
- Black carbon is said to be one of the largest contributors to climate change after CO₂. But unlike CO₂, which can stay in the atmosphere for years together, black carbon is short-lived and remains only for days to weeks before it descends as rain or snow.

Black Carbon (Soot)

BLACK CARBON (BC)

SOURCES

Black carbon is one of many particles and gases that are emitted when diesel, coal, and other biomass fuels are burned.



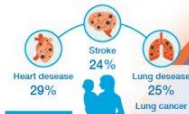
Black carbon, or soot, is part of fine particulate air pollution ($PM_{2.5}$) and contributes to climate change.

IMPACTS

CLIMATE

Absorbs sunlight and converts it to heat

Deaths from



HEALTH

7 million pollution-related deaths every year

LIFETIME IN ATMOSPHERE: UP TO 2 WEEKS

Since black carbon does not last long in the atmosphere, efforts to reduce it will bring immediate benefits for the climate and human health.

WEATHER

- Prevents clouds from being formed
- Alters regional weather patterns and rainfall

SNOW & ICE

Accelerates the melting of snow and ice

AGRICULTURE & ECOSYSTEMS

- Reduces sunlight
- Affects plant health and productivity

Study finds increased concentration of black carbon in Gangotri region during tourist seasons

A study finds remarkable increase in concentration of black carbon in the atmosphere near the pilgrim town of Gangotri in Uttarakhand during two annual tourist seasons

Black Carbon (Soot) and COVID-19



According to a study conducted by the **Indian Institute of Tropical Meteorology**:

- The COVID-19 infection count increase corresponded with an increase in black carbon in the atmosphere and decreased simultaneously with the decrease in black carbon in the atmosphere.
- The COVID-19 death count increased with an increase in black carbon in the atmosphere and decreased when there was a decrease in black carbon in the atmosphere.

Read at: <https://www.downtoearth.org.in/blog/air/black-carbon-particles-contribute-to-increasing-covid-19-cases-78130>

Remember : prelims question : vectors for virus are biological only. (wrong)

Chlorofluorocarbons (CFCs)

- CFCs were phased out via the Montreal Protocol due to their part in ozone depletion. These anthropogenic compounds are also GHGs, with a much higher potential to enhance the greenhouse effect than CO₂.

Hydrofluorocarbons (HFCs)

- HFCs are used as refrigerants, aerosol propellants, solvents, and fire retardants. These chemicals were developed as a replacement for CFCs. Sadly, HFCs are potent GHGs with long atmospheric lifetimes.

Fluorinated Gases



HYDROFLUOROCARBONS (HFCs)

HFCs are powerful man-made greenhouse gases that are rapidly building up in the atmosphere.

CONSUMPTION

HFCs are a group of industrial chemicals primarily used for air conditioning and refrigeration.

RESIDENTIAL, COMMERCIAL AND INDUSTRIAL AIR CONDITIONING AND REFRIGERATION

47%



HFC consumption today...



- filling new equipment
- topping up leaking equipment

Up to 10 air conditioners will be sold every second over the next 30 years

% = global emissions

LIFETIME IN ATMOSPHERE:

15 YEARS (AVERAGE WEIGHTED BY USE)

Many HFCs are short-lived climate pollutants. The most abundant of these, HFC-134a, is 3,790 times more damaging to the climate than carbon dioxide (CO₂) over a 20-year period.

MOBILE AIR CONDITIONING

24%



FOAM AGENTS

11%



UNITARY AIR CONDITIONING

8%



AEROSOLS

5%



FIRE EXTINGUISHERS AND SOLVENTS

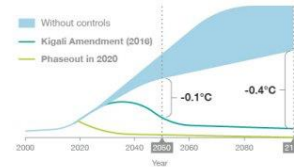
5%



AVOIDING FUTURE EMISSIONS

HFC emissions will quickly grow without action. Under the Kigali Amendment, countries have committed to reduce the production and use of these gases by 85% by 2050.

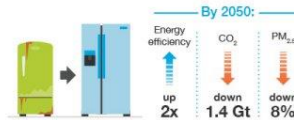
This would avoid up to 0.1°C of warming by 2050 and up to 0.4°C by 2100.



ADDED BENEFITS OF ENERGY EFFICIENCY

Almost 80% of the climate impact from cooling comes from the electricity generated to power the equipment.

Improving air conditioning efficiency could cut CO₂ and air pollutant emissions from power generation and as much as double the climate benefit of the Kigali Amendment.



www.ccacoalition.org/hfc

Perfluorocarbons or Fluorocarbon (PFCs)

- PFCs are a group of human-made chemicals composed of carbon and fluorine only.
- They are produced as a by-product in aluminium production and the manufacturing of semiconductors (as alternatives to CFCs).
- Like HFCs, PFCs generally have long atmospheric lifetimes and high GWP.

Sulphur Hexafluoride (SF₆)

- Sulphur hexafluoride is also a GHG.
- It is used in magnesium processing and semiconductor manufacturing, as well as a tracer gas for leak detection.
- It is used in electrical transmission equipment, including circuit breakers.

Nitrous Oxide (N₂O)



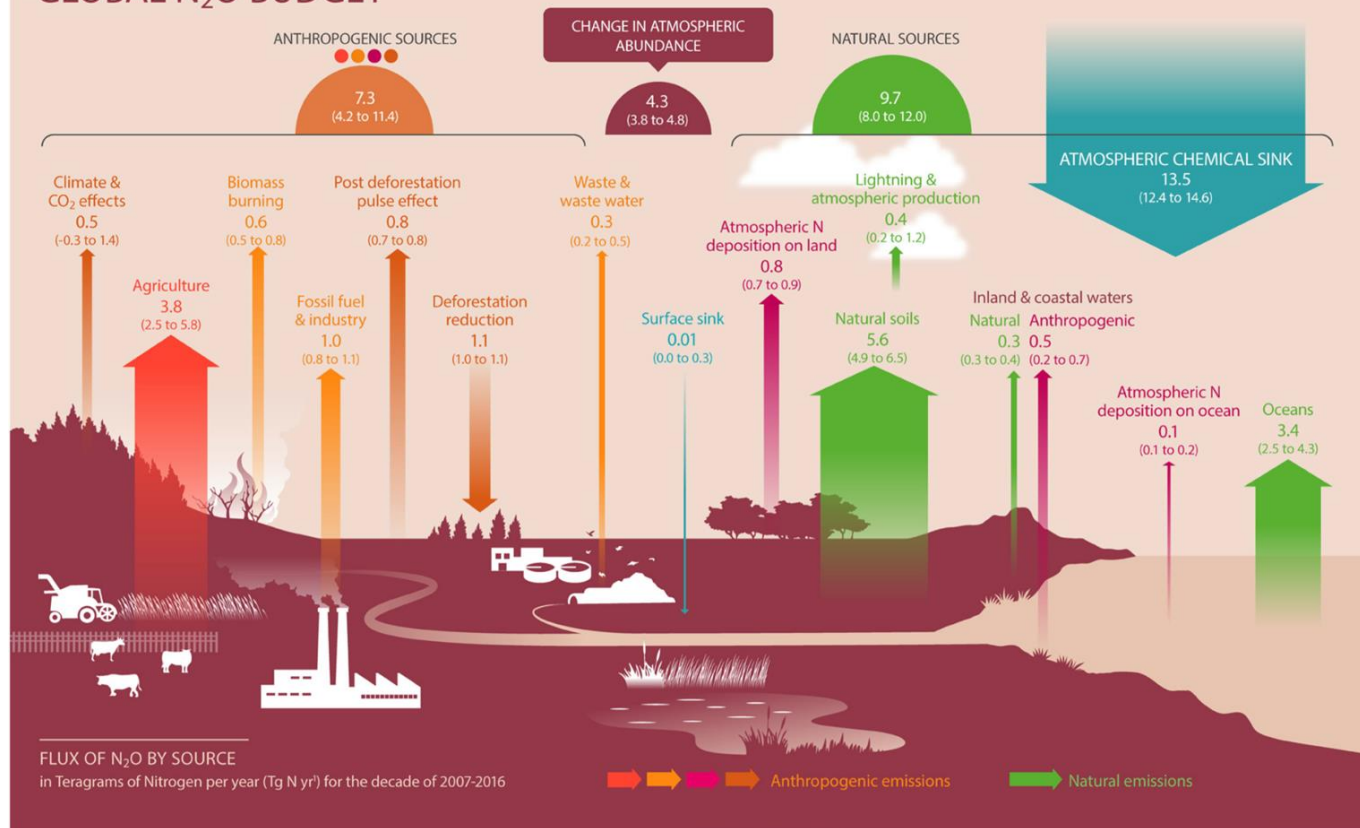
- **Nitrous Oxide (N₂O)**
- It is a GHG (300 times more potent than carbon). It is also the only remaining threat to the ozone layer as it can live in the atmosphere for up to 125 years (long-lived, just like CO₂).
- N₂O emissions have increased by 30 per cent between 1980 & 2016. It has the third-highest concentration after CO₂ and methane in our atmosphere among GHGs responsible for significant global warming.
- 43% of the total emissions of N₂O came from human sources, a significant portion of it from the agricultural sector (nitrogen-based fertilisers).
- Most N₂O emissions have come from emerging countries like India, China & Brazil. Agriculture accounts for over 70% of all nitrous oxide emissions in the Indian environment, of which fertilisers, mostly urea, contribute 77%

Nitrous Oxide (N₂O)



- **Nitrogen Oxide (NO), Nitrogen dioxide (NO₂) are global cooling gasses.**
- **Nitrous Oxide (N₂O) is a greenhouse gas.**
- **They all fall under the general category of oxides of nitrogen (NO_x)**

GLOBAL N₂O BUDGET



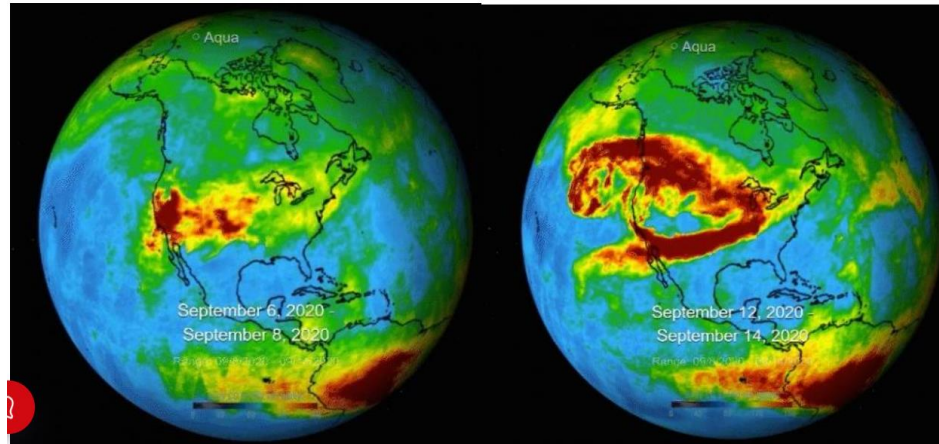
Carbon monoxide



- Carbon monoxide (less dense than air) is a short-lived, very weak direct GHG. Through natural processes in the atmosphere, it is eventually oxidised to carbon dioxide (GHG).
- It has an indirect radiative forcing effect by elevating concentrations of methane (GHG) and tropospheric ozone (GHG).

Carbon monoxide: In News

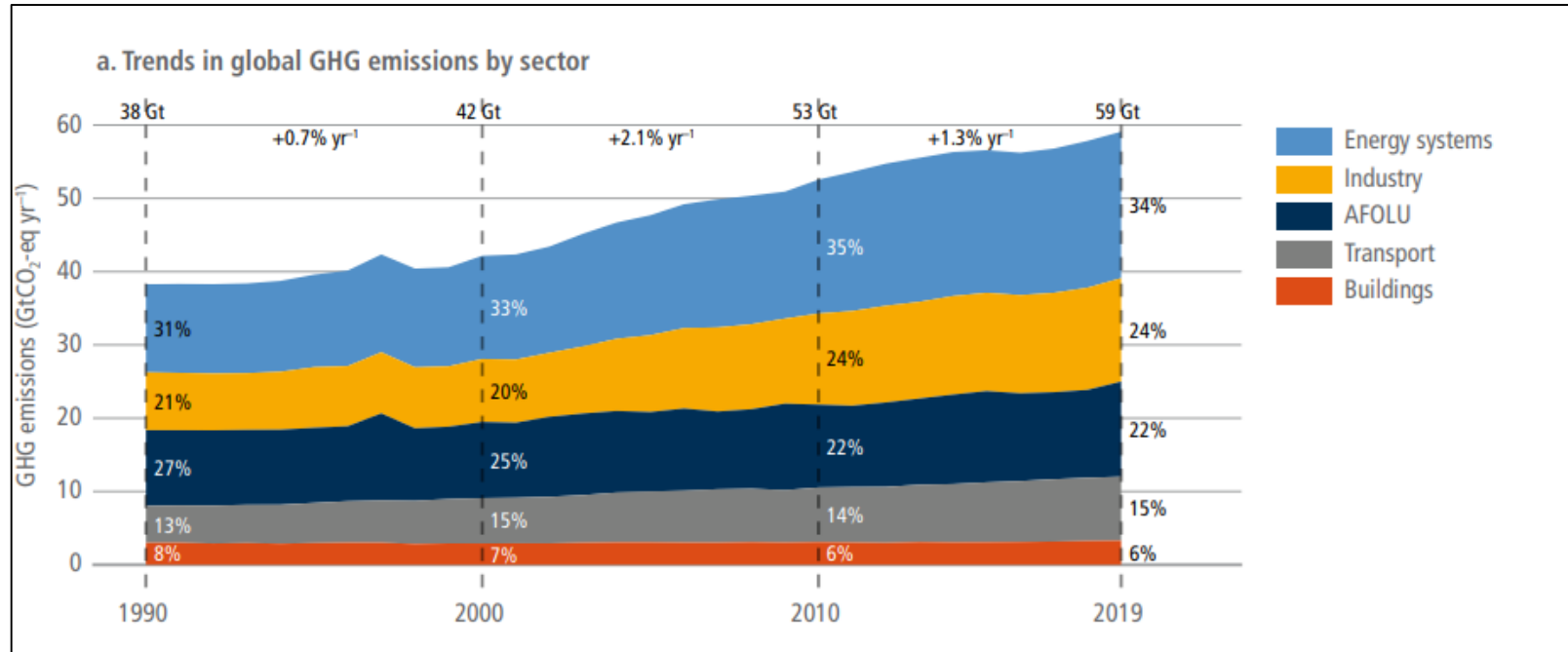
- Carbon monoxide plumes from the wildfires in California, United States spread across the other side of the country in the East coast and across the Great Lakes and Canada, showed data from NASA.
- The plumes were carried by the atmospheric jet stream that blew the carbon monoxide plume eastward across the US and over the Atlantic Ocean, according to the US National Weather Service.



- **Water Vapour**
- It is one of the most variable gaseous substances in the atmosphere – constituting between 0.02% and 4% of the total volume (in cold, dry and humid tropical climates, respectively).
- Water vapour is also a variable gas in the atmosphere, which decreases with altitude. 90% of moisture content in the atmosphere exists within 6 km of the earth's surface.
- Water vapour also decreases from the equator towards the poles.
- Like carbon dioxide, water vapour plays a significant role in the insulating action, of the atmosphere.
- It absorbs not only the long-wave terrestrial radiation (infrared radiation or heat emitted by the earth during the night), but also a part of the incoming short-wave solar radiation (visible and UV radiation).

- **Carbon Footprint**
- Carbon footprint is a measure of the impact of one's activities have on the amount of CO₂ produced through the burning of fossil fuels and is expressed as the weight of CO₂ emissions produced in tons.
- It is measured during the course of a year & can be associated with an individual, organisation, product etc.
- For example- India has pledged to reduction by 45% in the emissions intensity (carbon footprint) of its economy by 2030 compared to 2005 levels.

Greenhouse Gases

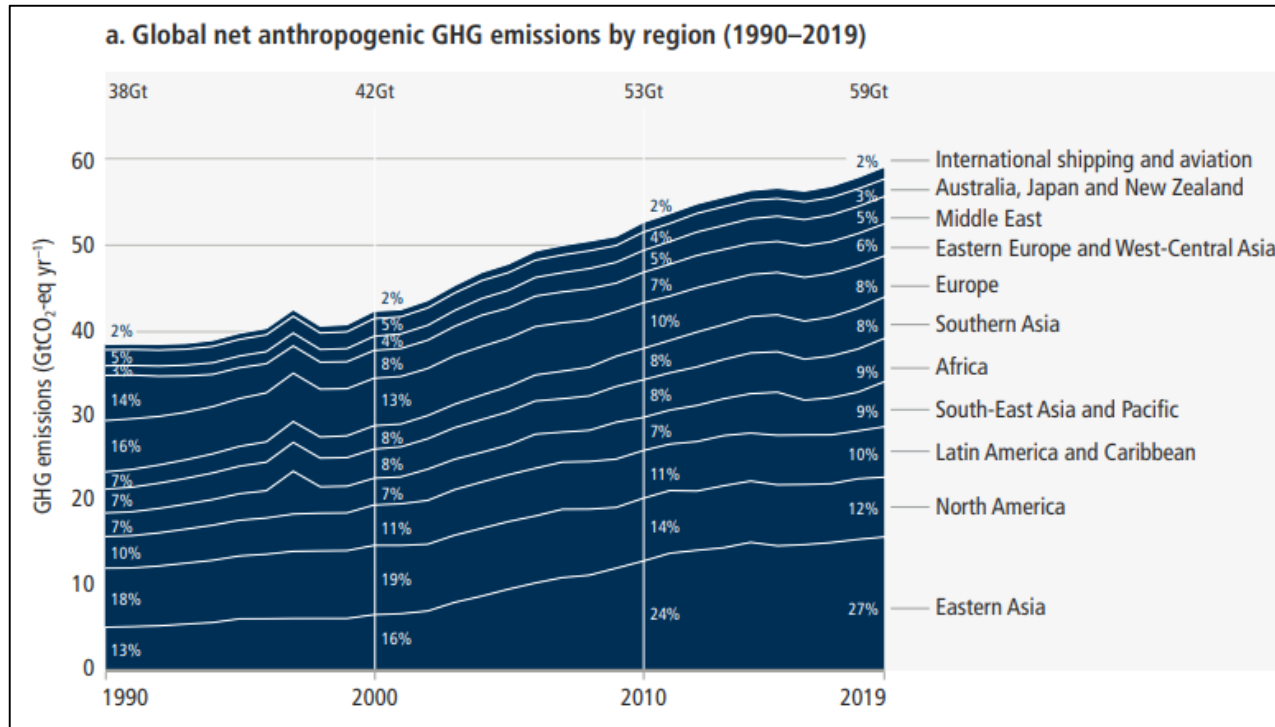


AFLOU- Agriculture, forestry and other land use (AFOLU)

Source-
https://www.ipcc.ch/report/ar6/wg3/downloads/report/IPCC_AR6_WGIII_Chapter02.pdf

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Greenhouse Gases



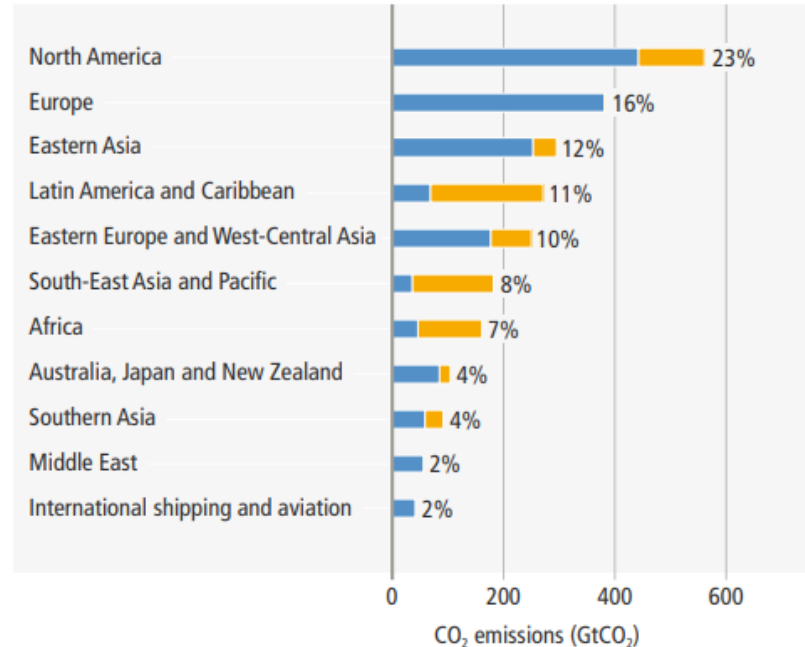
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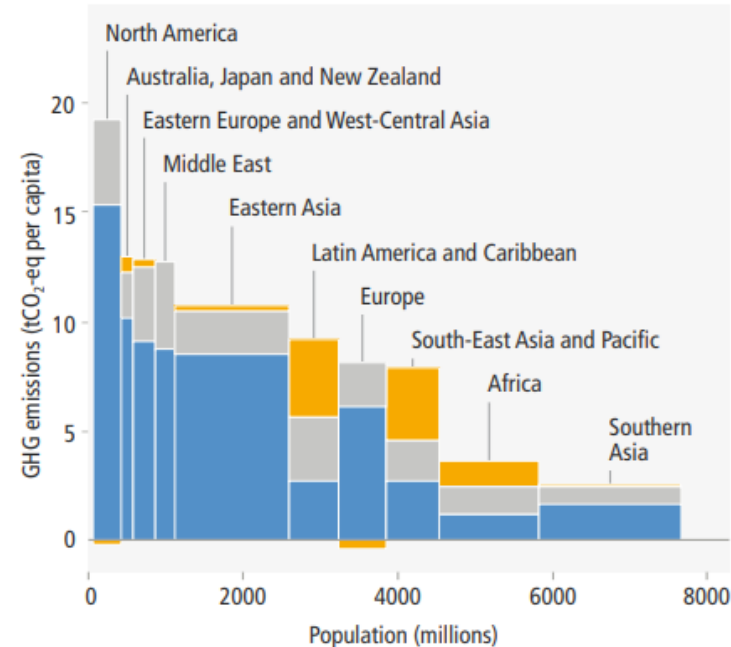
Greenhouse Gases



b. Historical cumulative net anthropogenic CO₂ emissions per region (1850–2019)



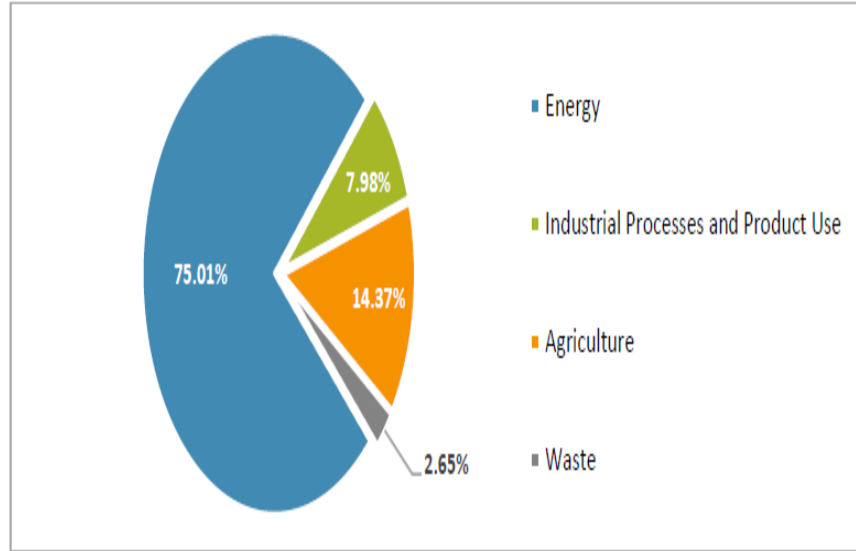
c. Net anthropogenic GHG emissions per capita and for total population, per region (2019)



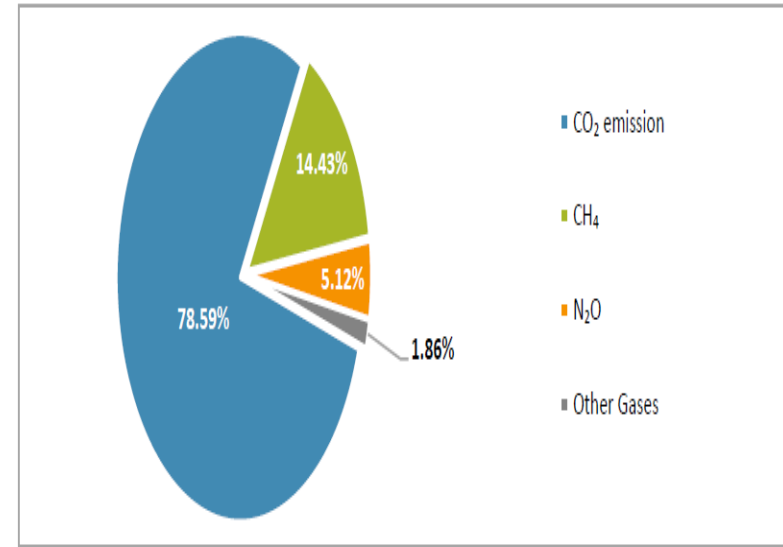
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https://www.ipcc.ch/report/ar6/wg3/downloads/report/IPCC_AR6_WGIII_SPM.pdf

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Greenhouse Gases



Emission by sector- India



Emission by GHG- India

Source-
3rd BUR
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- **Carbon Bomb**
- A Carbon Bomb is “a coal, oil or fossil gas project with a potential to emit over a Gigaton of CO₂ emissions over its lifetime.”
- A network named “Leave It In the Ground Initiative (LINGO)” has identified 425 of them around the world. According to LINGO, Carbon bombs' potential emissions exceed a 1.5°C carbon budget by a factor of two.
- China, the United States, Russia, & Saudi Arabia have the highest number of Carbon Bombs.
- Some listed carbon bomb projects include:
- Carmichael Coal Project (coal mine in Queensland, Australia & owned by the Adani Group)

Greenhouse Gases



- Gevra Coal Mines in Chhattisgarh by Coal India.
- Rajmahal Coal Mines in eastern Jharkhand is owned by Eastern Coalfields.
- **About LINGO**
- LINGO is a think tank with a mission to “leave fossil fuels in the ground & learn to live without them.”
- It envisions a world powered by 100% renewable energy & advocates a circular economy.
- LINGO aims to gain ground support for protesting such projects & challenge them through litigation.

- Carbon sensitivity refers to the sensitivity of the Earth's climate to changes in atmospheric carbon dioxide (CO₂) concentrations. It is a measure of how much the Earth's temperature would change in response to a certain increase or decrease in atmospheric CO₂.
- The concept of carbon sensitivity is usually expressed in terms of "climate sensitivity," which is defined as the equilibrium global mean surface temperature change that would result from a doubling of atmospheric CO₂ concentrations above pre-industrial levels.
- This value is typically estimated to be between 1.5°C and 4.5°C, with a most likely value of 3°C.

- Climate feedback refers to the interactions between the various components of the Earth's climate system that either amplify or dampen the effects of an initial climate forcing, such as an increase in greenhouse gas concentrations. These feedback mechanisms can either accelerate or slow down climate change, and are an important factor in determining the overall sensitivity of the climate system to external perturbations.
- One of the most well-known examples of a positive climate feedback is the ice-albedo feedback. As global temperatures rise, ice and snow melt, exposing darker surfaces that absorb more solar radiation, further warming the Earth and causing more ice and snow to melt. This feedback loop amplifies the initial warming, leading to even more ice loss and warming.

- On the other hand, negative feedback mechanisms can act to reduce the effects of a climate forcing. For example, as atmospheric CO₂ concentrations increase, plants may grow more rapidly, absorbing more carbon and reducing the atmospheric CO₂ concentration.
- Other examples of climate feedbacks include changes in cloud cover, ocean circulation, and the release of greenhouse gases from permafrost or ocean sediments.
- Permafrost melts : earth releases stores of greenhouse gases from melting permafrost and the temperature rises, creating a positive feedback.
- Water vapour : world warms, water vapour in atmosphere increases and so does the temperature.

Effect of Clouds on Climate Change



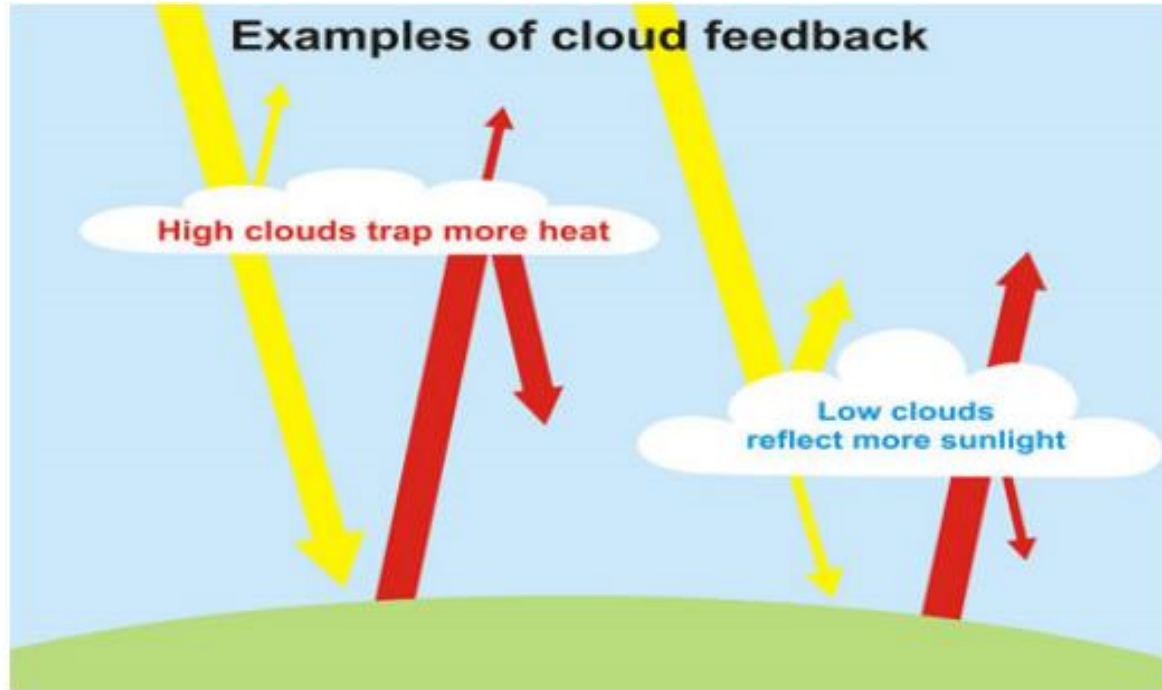
- Clouds have a dual effect on our climate,
 - 1) they reflect part of the sunlight back into space, which decreases temperatures (negative feedback), and
 - 2) they trap part of the heat reflected from the earth's surface, which increases temperatures (positive feedback). The net effect depends on the type of cloud.

Effect of Clouds on Climate Change



- A warmer and wetter atmosphere will affect cloud cover. Clouds that contain more water droplets are optically thicker and more effective at blocking sunlight than those composed mainly of ice crystals (cirrus clouds).
- Low clouds are “strong coolers” since they reflect a large part of solar radiation (sunlight) and absorb little heat from the ocean and land. On the other hand, very thin, high clouds such as cirrus clouds reflect little sunlight, but they are very efficient at absorbing thermal radiation (heat), making them “strong warmers”.

Effect of Clouds on Climate Change



Effect of Clouds on Climate Change



- Meanwhile, a shift in sun-blocking clouds (due to global warming) from the tropics towards the poles, here the incoming sunlight is less intense, would decrease their power to block sunlight.
- All this means the global net effect of cloud feedback is complex and hard for scientists to model precisely.

Impact of Global warming



- **Increased Frequency and Severity of Heat Waves**
- A heat wave is a period of excessively hot weather. India Meteorological Department (IMD) declares heat waves when the actual maximum temperature remains 45°C or more, regardless of the normal maximum temperature.
- Example- British Columbia (South-Western Canada) & neighbouring Washington and Oregon states (North-Eastern US) in 2021, Australia (2019-20), California (2020), and Siberia.
- Heat waves are caused due to shifting of Jet Streams (meandering Rossby Waves in the temperate region cause Heat Domes), hot local winds like loo (affects Gangetic Plains Region), and anthropogenic causes like global warming.
- **read from Geography

Impact of Global warming



- Communities close to coastal environments, small island nations, polar areas and high mountains are particularly vulnerable to changes, such as rising sea levels and shrinking glaciers.
- Communities in other areas are affected by extreme weather events exacerbated by ocean warming
- Statistics
- Marine heat waves have become twice more frequent in the past four decades and are lasting longer.
- The report finds that human activities are responsible for 84 to 90 per cent of the marine heat waves that occurred in the last decade.
- By 2081, the frequency of marine heat waves could jump by 20 to 50 times.

Impact of Global warming



- **Marine Heat Waves**
- Marine heatwaves occur when ocean temperatures for a particular oceanic location are unusually extremely warm for an extended period and time of year.
- They can occur in summer or winter
- IPCC Special Report on the Ocean and Cryosphere in a Changing Climate (SROCC) on Marine Heat Waves
- Today's oceans are experiencing unprecedented conditions with increased temperatures, further ocean acidification, marine heatwaves and more frequent extreme El Niño and La Niña events.

Impact of Global warming



- Impact of marine heatwaves
- **Marine productivity**
- They have resulted in large-scale coral bleaching, which takes more than 15 years for corals to recover from.
- Marine heat waves reduce the mixing between water layers and, consequently, the supply of oxygen and nutrients for marine life.
- The Pacific Ocean, which had seen such unusually warm water, had boosted the growth of toxin-producing algae and suppressed the growth of small organisms at the base of the ocean food chain.

Impact of Global warming



- **Impact on weather patterns**
- A more pronounced effect of marine heat waves would be on global wind circulation and ocean currents.
- IPCC report indicates, the Atlantic Meridional Overturning Circulation (which ensures a northward flow of warm, salty water in the upper layers of the Atlantic and a southward flow of colder, deep waters) has already weakened which could have cascading effects.
Please READ AMOC from geography.
- **More severe cyclonic storms**
- IPCC says there is emerging evidence of an annual increase in the percentage of category 4 and 5 storms.
- These storms sustain their strength further by feeding on the moisture over warm ocean waters.

Impact of Global warming

Marine Heatwaves occur everywhere in the ocean

2003: Mediterranean Sea

4°C warmer than average for 30 days
Largest event on record
Mass mortality of marine life in rocky reefs

Warm air ("normal heatwaves")
can drive marine heatwaves by
warming the ocean surface

Ocean currents can drive
marine heatwaves by moving
around warm water

Climate modes, like El Niño, can cause
marine heatwave events to occur

2013-2015: "The Blob"

2½°C warmer than average for 226 days
Longest event on record
Caused unseasonably warm weather in
Pacific Northwest of USA and Canada

2012: Northwest Atlantic

2½°C warmer than average for 56 days
Largest event on record
Lobster fishery peaked early and led to
Canada-USA economic tensions

2011: Western Australia

Over 3°C warmer than average for 60 days
Largest event on record
Seaweeds, fish and sharks moved south

Impact of Global warming



- **Urban Heat Islands**
- An urban heat island is an urban or an industrial area that has temperatures considerably higher than its surrounding rural areas (both share the same climate) primarily due to anthropogenic reasons.
- Causes
- Heat-trapping concrete and asphalt (have very low albedo or low reflectivity) replace natural vegetation and water (reduced evaporation and evapotranspiration).
- High-rise buildings that offer more surface area for heat absorption.
- High vehicle density and heat released from their engines.

Impact of Global warming



- High pollution levels and high concentrations of GHGs like CO₂ (thermal power plants and vehicles).
- GHGs, aerosols, particulate matter, etc., are good at absorbing outgoing infrared radiation.
- Cooling appliances like air conditioners release heat into the surroundings.
- Bad monsoons because of less evaporation of water from vegetation (evapotranspiration) and soil.
- Heat islands at night: Cities used to cool down during the night, but nowadays, they are trapping this heat overnight. ACs, pollution and a close & dense networks of buildings are the primary reason for the Urban heat islands at night.

Impact of Global warming



- Albedo of a surface is the proportion of sunlight that the surface can reflect back into space.
- Fresh snow has the highest albedo (reflects back 80-90% of the incident sunlight). Dark asphalt used to road construction and in the roofing of buildings has very low albedo (~5%).

Impact of Global warming

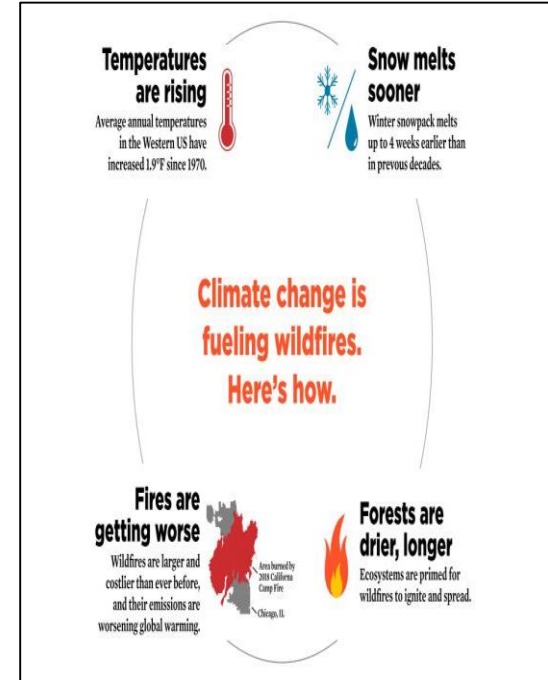


Surface	Albedo
1. Fresh Snow	80-90%
2. Thick cloud (low level clouds)	70-80%
3. Water near horizon	50-80%
4. Old Snow	45-50%
5. Desert	30-45%
6. Light soil	20-45%
7. Thin cloud (Cirrus clouds)	25-35%
8. Grasses	20-25%
9. Soil	20-25%
10. Crops	10-25%
11. Forest	10-20%
Asphalt (used in road construction & roofing)	5%

Impact of Global warming



- **Increased Incidence of Wildfires**
- Increased incidence of wildfire creates a positive feedback loop exacerbating global warming.
- In climate change, a feedback loop is something that speeds up or slows down a warming trend. A positive feedback accelerates a temperature rise, whereas a negative feedback slows it down.
- Example- Australia's Bushfires are Getting Severe



Impact of Global warming



- **Wildfires in Arctic**
- Wildfires on permafrost in Siberia south of the Arctic are not uncommon.
- But in 2020, burning occurred well above the Arctic Circle (tundra), a region not commonly known to support large wildfires.
- Reason- unprecedented drying up of the tundra vegetation like moss, grass & dwarf shrubs.
- ‘Zombie fires’ are becoming more frequent in the once-frozen tundra — north of the Arctic Circle.
- A zombie fire (holdover fire) is a fire from a previous growing season that can burn slowly with smoke but no flame under the ground made of carbon-rich peat.
- Impact- The fires & record temperatures have the potential of turning the carbon sink into a carbon source.

Impact of Global warming



- **Shrinking Cryosphere**
- The areas of snow or ice, which are subject to temperatures below 0°C for at least a part of the year, compose the cryosphere.
- Continental ice sheets found in Greenland and Antarctica, ice caps, glaciers, areas of snow (glaciers of Alps, Himalayas) and permafrost (Siberia), frozen parts of the ocean, rivers, lakes, etc., are all part of the cryosphere.
- Glaciers from almost half of natural World Heritage sites- such as the Khumbu Glacier in the Himalayas- may disappear entirely by 2100, if emissions continue at the current rate, an IUCN study warns.
- The study predicts glacier extinction in 21 of the 46 natural World Heritage sites with glaciers.

Impact of Global warming



- **Importance of cryosphere**
- Snow and ice (have the highest albedo) reflect heat from the sun (heat budget). (refer earlier slide)
- Supply freshwater to many parts of the world
- Cryosphere is most sensitive to climate shifts. Hence, it acts as the earth's black box (ice accumulates layer over layer) > Studying the vertical ice column helps understand past global changes in climate

Impact of Global warming



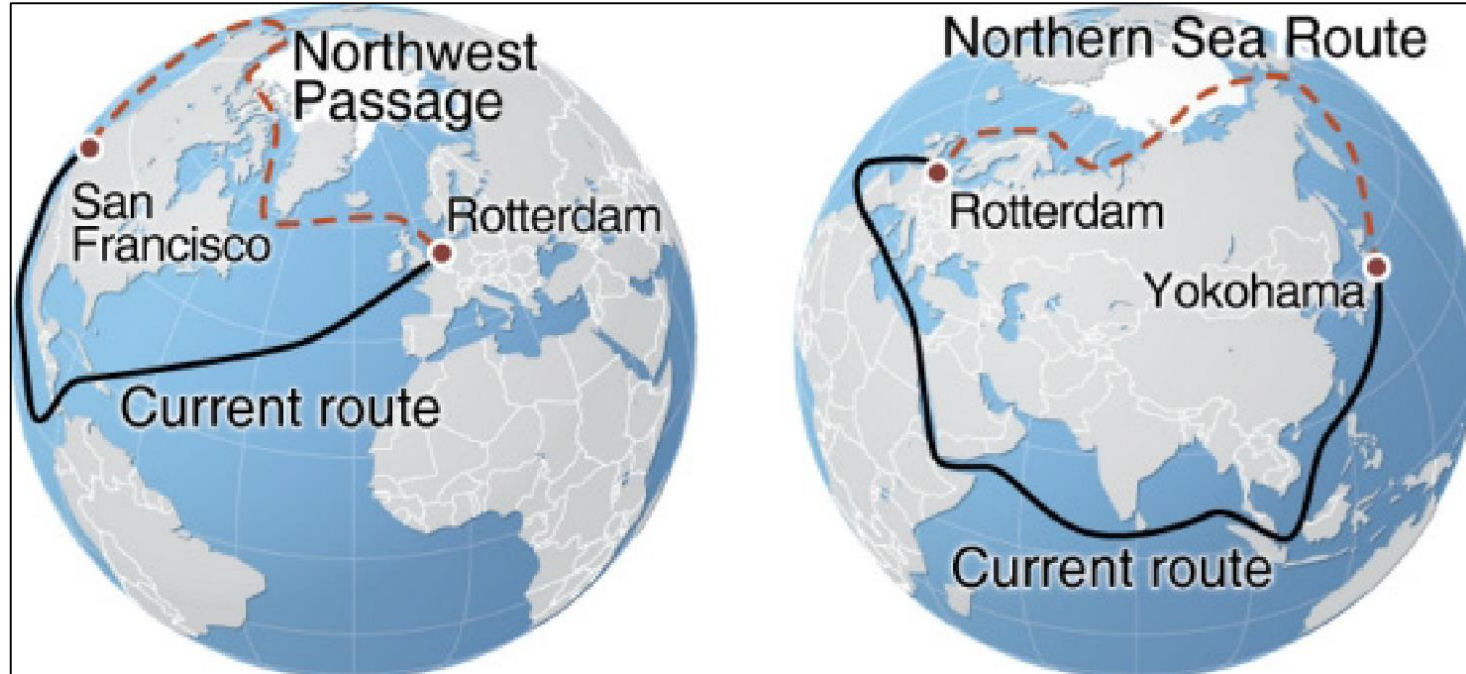
- **Impact of shrinking cryosphere**
- Water scarcity and water wars between nations.
- Loss of ecologically rich and important coastal wetlands.
- Distress migration of coastal inhabitants due to the submergence of major coastal cities.
- Submergence of Small Island Developing States (first to suffer the consequences of climate change).
- Reduced hydroelectric power generation due to the abnormal behaviour of glaciers will further increase dependence on fossil fuels.

Impact of Global warming



- Vegetation change- The tundra will return to swamps (> loss of forest > loss of carbon sink), the permafrost is thawing (> exposing the subsurface carbon sinks).
- The Northern Sea Route (NSR) which would connect the North Atlantic to the North Pacific through a short polar arc is slowly opening due to ice melting. Models predict that this route could be ice-free in summer by 2050.
- Coastal flooding and reduction of arable land on coasts.

Impact of Global warming



Impact of Global warming



- The distance from Rotterdam to Yokohama will be cut by 40 per cent compared to the Suez route.
- Unexplored oil & natural gas deposits are estimated to be 22% of the world's unexplored resources, mostly in the Arctic Ocean (Barents Sea Region).
- Mineral deposits including 25 per cent of the global reserves of rare earths are buried in Greenland
- **Issues**
- Environmental risks.
- Unlike Antarctica, the Arctic is not a global common.
- There is no treaty that governs it, only the UN Convention of Law of the Sea (UNCLOS) deals it.

Impact of Global warming

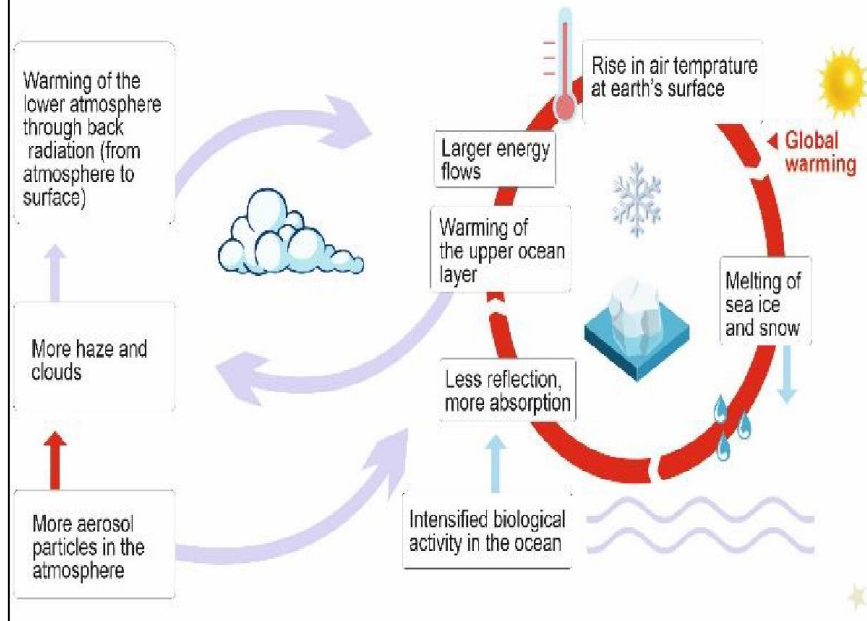


- **Arctic Amplification**
- Arctic is heating up 4 times as fast as rest of world due to Arctic amplification (AA).
- It refers to enhancement of near-surface air temperature change over Arctic relative to lower latitudes.
- Whenever there is change in net radiation balance, it is slightly higher in the arctic as compared to tropical areas.
- Reasons
- Change in Albedo:
- Amplification is primarily caused by melting polar ice, which is increasing in the Arctic at a rate of 13% per decade.
- Ice is more reflective of sunlight (high albedo) than land or ocean.

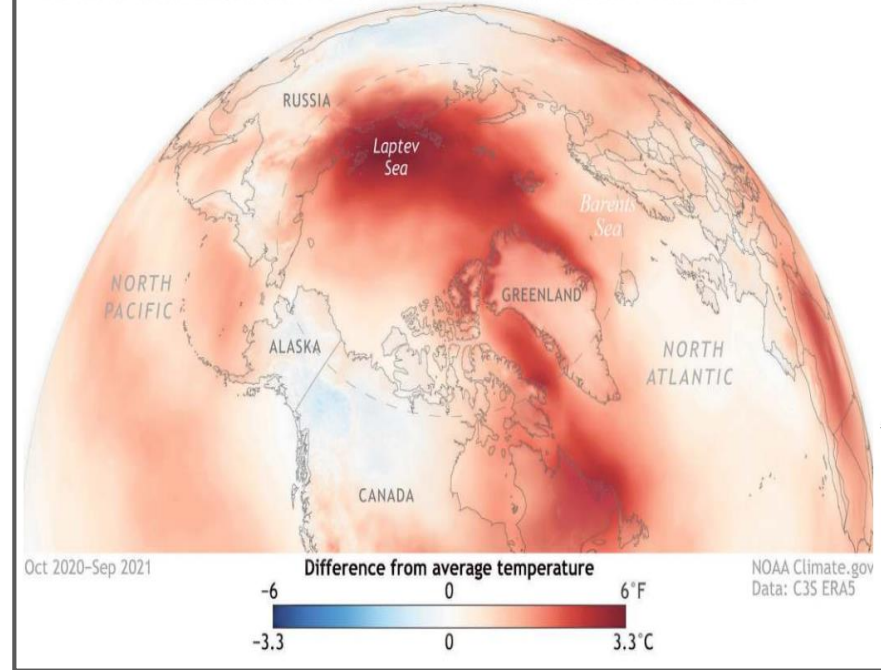
Impact of Global warming

Arctic Amplification

The dynamics of global warming



2021 WAS THE ARCTIC'S SEVENTH WARMEST YEAR ON RECORD



Impact of Global warming



- When ice melts, it typically reveals darker areas, & this results in increased sunlight absorption (low albedo) & associated warming.
- Melting sea ice also releases greenhouse gases from thawing permafrost & frozen methane from the ocean bottom & further intensifies Amplification.
- Polar amplification stronger in Arctic than Antarctic as:
- Arctic is an ocean covered by sea ice, while Antarctica is an elevated continent covered in more permanent ice & snow. In fact, the Antarctic continent has not warmed in the past seven decades, despite a steady increase in the atmospheric concentrations of GHGs.

Impact of Global warming



- **Impact of arctic amplification**
- Sea-level rise globally.
- Zombie ice from massive Greenland ice sheet will eventually raise global sea level by at least 10 inches.
- (Zombie ice- It's the ice that is still attached to thicker areas of ice, but is no longer getting fed by those larger glaciers because parent glaciers are getting less replenishing snow)
- Arctic permafrost thaw releases carbon dioxide & methane, which has the potential to cause even further warming (positive feedback).

Impact of Global warming



- **Impact of arctic amplification**
- Impacts on mid-latitude climate, as well as the occurrence of extreme events by:
- Weakening of the tropospheric Jet Stream (will contribute to an increase in unusual & extreme weather in the mid-latitudes).
- Weakening of the stratospheric polar vortex (which results in extreme weather events in the mid-latitudes).
- **Arctic Report Card 2022** (by US National Oceanic and Atmospheric Administration (NOAA), annually since 2006)
- **Key highlights of Report:**
- Arctic continues to warm more than twice as fast as rest of globe.
- Persistent summer sea ice due to cooler surface waters and north winds at Chukchi Sea.

Mains questions.



- How does the cryosphere affect global climate? (2017)
- Why is India taking keen interest in resources of Arctic Region? (2018)
- What is the economic significance of discovery of oil in the Arctic Sea & its possible environmental consequences? (2015)
- How do the melting of the Arctic ice and glaciers of the Antarctic differently affect the weather patterns and human activities on the Earth? Explain. (2021)

Sea Level Change



- **Sea level change** means the fluctuations in the mean sea level over a considerably long period of time. Commonly, seasonal variations of 5-6 cm are observed in a year. Sea level change means the fluctuations in the mean sea level over a considerably long period of time.

Processes that cause Change in Sea Level



Eustatic changes

- It occurs when the volume of sea water changes due to factors such as global warming and melting of ice sheets (rise in sea level) or ice ages (fall in sea level) and changes in the volume of mid-oceanic ridges.

Tectonic changes occur due to a change in land level

- Isostatic changes take place due to addition or removal of load — during ice ages, landmass subsided due to the load exerted by the glacial ice. On the other hand, landmasses rise as the glacial ice is removed.
- Epeirogenic movement occurs due to broad scale tilting of continents which may result in the rise of one part of the continent even as the other part may subside, causing an apparent rise in sea level.
- Orogenic movement (mountain building) results in the formation of lofty mountains and an apparent fall in sea level

Effects of Climate Change



Sea Level Rise and Coastal Flooding

- IPCC report warns that sea level rise could reach 60 to 110 cm if emissions continue to increase strongly.
- More than half the global population live in cities, most of which are located on low-lying islands and coasts & economic losses due to extreme flooding will soar 166 times more by 2050.

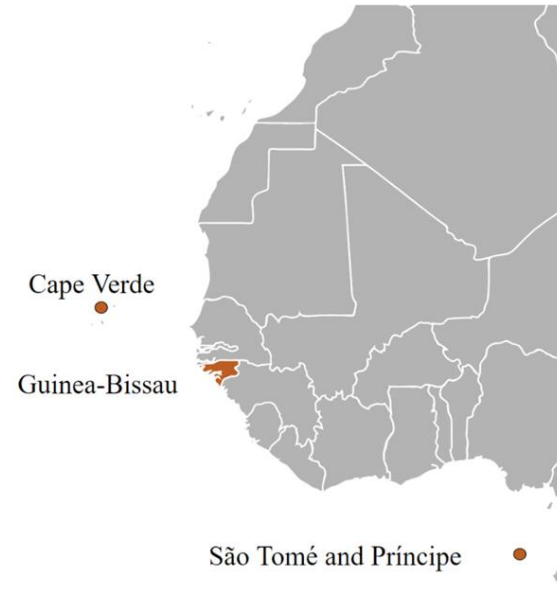


Effects of Climate Change

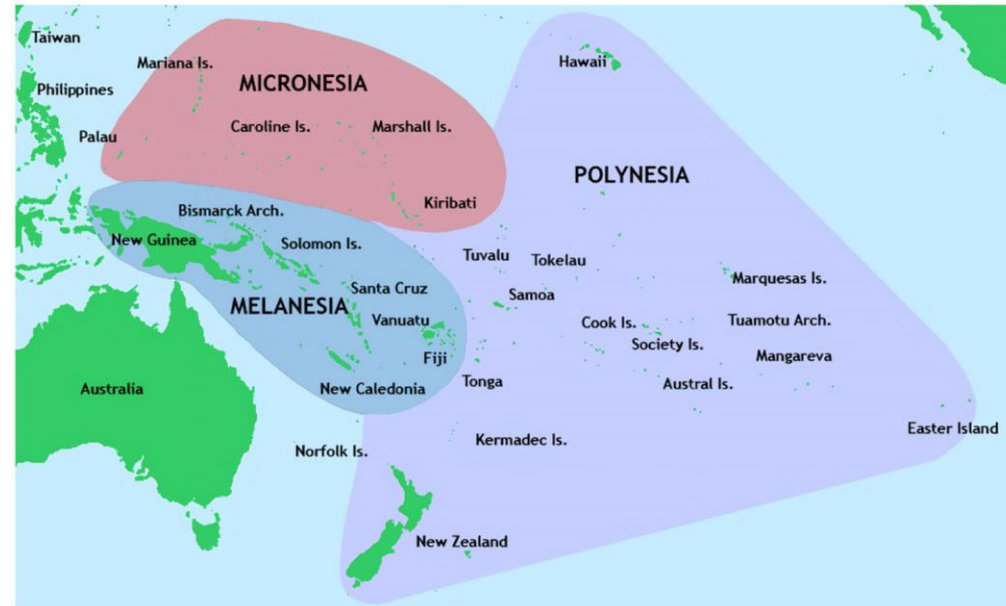
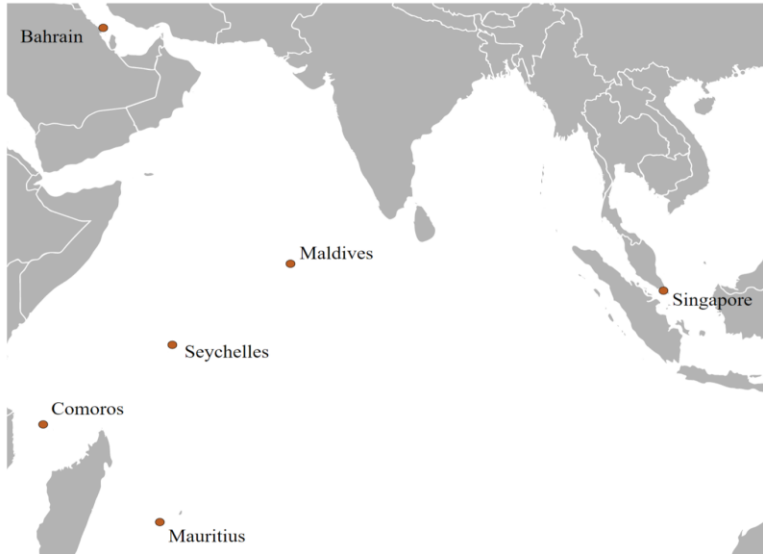


- Biggest losers here are small island development states or SIDS.
- They are most vulnerable to climate change.
- At the UN conference on environment and development (1992 Earth Summit) they were recognised as a distinct group of developing countries.
- Most of them are coral islands on shallow atolls.
- UN programme of action on sustainable development of SIDS is popularly referred to as the Barbados Plan of Action (BPOA). It is the only internationally approved programme specific to SIDS. (1994)
- Mauritius Strategy : 10 year comprehensive review of the BPOA in 2005.

Effects of Climate Change



Effects of Climate Change



Tropical Cyclones

- Tropical cyclones require a sea surface temperature (SST) of 26.5°C to form, while the highest intensity storms require much warmer sea surface temperature (SST) of $28-29^{\circ}\text{C}$. The frequent high intensity storms have been tied to the very warm SST ($30^{\circ}\text{C}+$).
- Regions further from the equator are more regularly experiencing the threshold temperature or $24-26^{\circ}\text{C}$ widening the range of formation of tropical cyclones. These conditions are exacerbated by global forcing mechanisms including El Niño, Indian Ocean Dipole, Southern Annular Mode and Madden-Julian Oscillation (these in turn are affected by global warming).

Tropical Cyclones

- The Arabian Sea is heating rapidly and driving more cyclones, and excessive rainfall (excessive rainfall in sea less moisture in monsoon winds less rainfall on the mainland). Climate models suggest that 64 per cent of the cyclone risk in the Arabian Sea was due to climate change.
- Unusual timing of cyclones in the arabian sea. Ex : cyclone vayu in june 2019.
- Changing path of cyclones in the arabian sea. Ex : Tropical Cyclones restricted to Gujarat earlier, but now affect kerala and karnataka too.

Deterioration of Carbon Sinks.

- High latitude forests store more carbon than tropical rainforests — one-third of the world's soil-bound carbon is in taiga and tundra areas. When the permafrost melts due to global warming, they release carbon in the form of carbon dioxide and methane.
- In the 1970s, the tundra was a carbon sink (takes more than it gives), but today, it is a carbon source, all because of global warming. (Global warming leads to more global warming positive feedback loop).

Carbon Dioxide Fertilization

- Earth's vegetated lands have shown significant greening (an increase in leaves on plants/trees) largely due to rising levels of atmospheric CO₂ (increases photosynthesis).
- Carbon dioxide fertilization contributes to 70 percent of the greening effect. The second most important driver is nitrogen, at 9 percent. The rest occurs due to land cover changes, precipitation, sunlight changes, etc.
- Plants acclimatize to rising CO₂ concentration and the fertilization effect diminishes over time. That is, raising CO₂ concentrations may be beneficial in the short run, but in the long-run it is harmful due to climate change.
- Therefore, carbon fertilisation is increasing carbon sink on land temporarily.

Climate Migrants

- Environmental Migrants are people displaced due to adverse changes to their local environment. Climate migrants are migrants people displaced due to climate change impacts such as sea level rise (Sundarbans), floods (Ganges, Brahmaputra basins), drought (central India, Vidarbha, Telangana, Rayalaseema), etc.
- The 64th round of NSSO, in 2007-08, titled *Migration in India* identified natural disasters including floods and droughts as one of the major reasons for migration

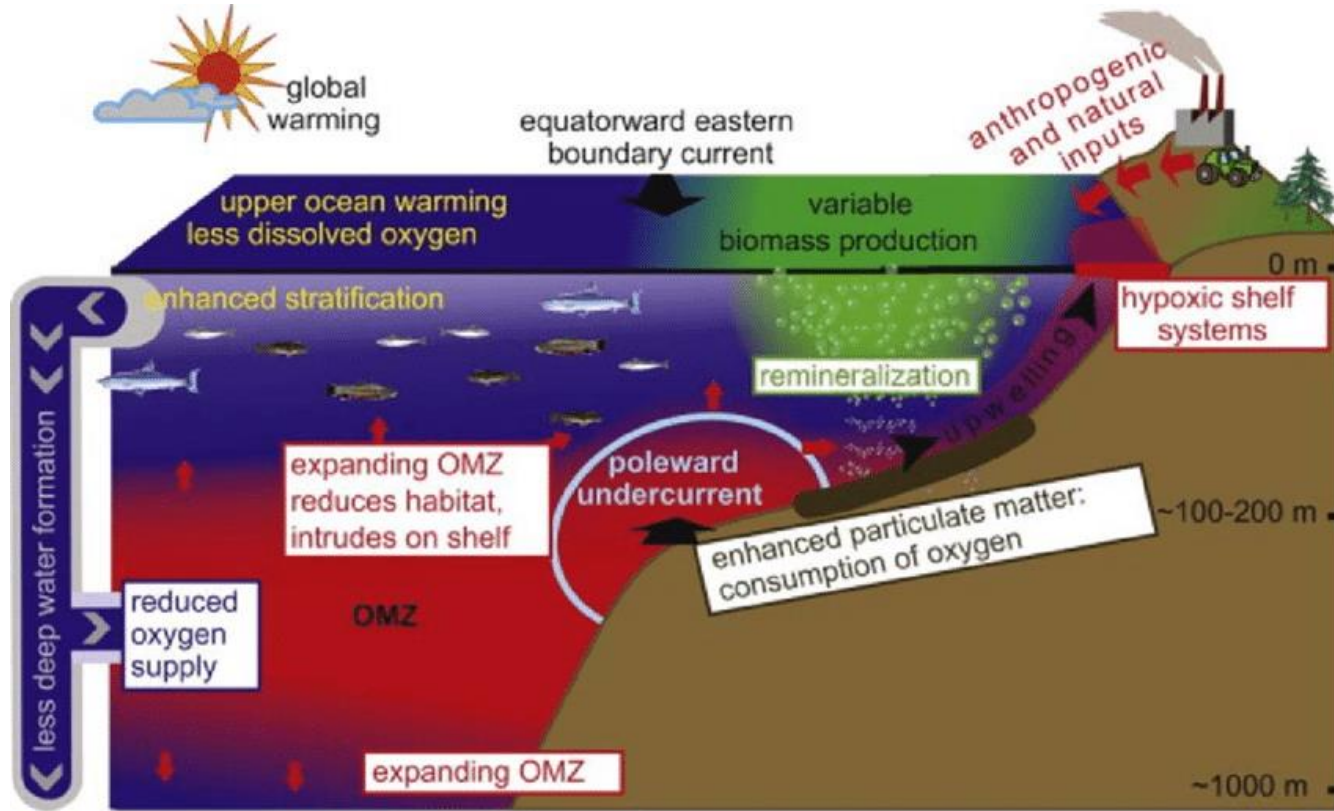


- **Economic Losses**
- Economic losses suffered involve money spent on adaptation to climate change (costs involved in moving from submerging areas to higher ground), rebuilding post extreme climatic event and money spent on climate change mitigation (like carbon sequestration).
- The economic losses suffered due to the emission of one ton of carbon dioxide into the atmosphere is termed the social cost of carbon. It is expressed as the dollar value of the losses suffered.
- India's country level social cost of carbon emission was estimated to be the highest at \$86 per ton of CO₂.
- It means the Indian economy will lose \$86 by emitting each additional ton of CO₂. India is followed by the US, where the economic damages would be \$48 per ton of CO₂ emissions.

- **Ocean Deoxygenation**

- Ocean deoxygenation is the expansion of oxygen minimum zones (OMZs) in the world's oceans as a consequence of anthropogenic emissions of carbon dioxide.
- OMZs are found in areas where an interplay of physical (ocean stratification) and biological (less photosynthesis) processes create a “pool” of water where oxygen concentrations fall (anoxic zones).
- Warmer oceans cause deoxygenation both because oxygen is less soluble in warmer water, and through temperature-driven stratification.
- The oxygen minimum zone (OMZ), sometimes referred to as the shadow zone, is the zone in which oxygen saturation in seawater in the ocean is at its lowest.

Effects of Climate Change



Food and Health Security at Risk

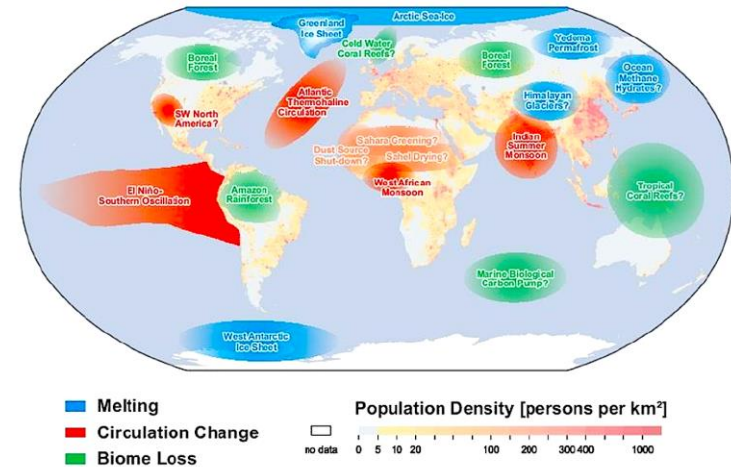
- Climate Change affects crops by impacting irrigation, insolation as well as the prevalence of pests. Increased frequencies of droughts, floods, storms and cyclones are likely to increase agricultural production variability.
- Rising temperature would increase fertilizer requirement and result in higher GHG emissions, **ammonia volatilization** (Ammonia volatilization results in a net loss of nitrogen from the soil system) and cost of crop production.
- Moderate warming (increase of 1 to 3°C in mean temperature) is expected to benefit crop yields in temperate regions, while in lower latitudes the crops will take a hit. However, the natural calamities due to global warming can offset the benefits in temperate regions.
- Spread of diseases (like malaria, etc.) in the tropics will put more pressure on the healthcare sector.

Climate Tipping Points

- Climate tipping points are points where any small change could push the earth system into abrupt or irreversible change.
- A new research highlighted that Antarctica is approaching a climate tipping point by 2060-when Antarctic ice could begin melting at a runaway rate.
- Globally 9 climate tipping points including the Antarctic Ice Sheet

Tipping elements in the climate system

Revised from original in Lenton et al. (2008) PNAS 105(6): 1786-1793



TIPPING POINTS
➔ CONNECTIVITY

- Click Here to Learn More**

- Coral reefs are made up of calcareous skeletons of thousands of tiny marine organisms called coral polyps.
- **Phylum:** Cnidaria (Related to anemones and jellyfish (cnidarians)).
- Colors differs wrt the nature of the salts they are made of.
- Polyps are shallow warm-water organisms with soft bodies covered by calcareous skeletons. The polyps extract calcium salts from seawater to form these hard, tubular skeletons. Small marine plants (algae) also deposit calcium carbonate contributing to coral growth.

Corals

- Polyps live in colonies on the seafloor as a cemented calcareous rocky mass, collectively called corals.
- When the coral polyps die, they shed their skeleton (coral) on which new polyps grow.
- The cycle is repeated for millions of years, leading to the accumulation of layers of corals.
- The shallow layers of corals created by the depositions of corals are called coral reefs.



Types of Corals in News

- **Acropora corals**
- A recent report stated that highest levels of coral cover, within the past 36 years, have been recorded in northern and central parts of GBR, fuelled largely by increases in fast-growing Acropora corals.
- These fast-growing corals are also most susceptible to environmental pressures such as rising temperatures, cyclones, pollution etc.



Types of Corals in News

- **Black Corals (Deep Sea Corals)**
- Researchers have discovered new species of black corals living 2,500 feet below surface off the coast of Australia.
- They are all over the world and at all depths. However, typically, they are known as deep-sea corals.
- They are rarely black, but rather vary in color from white to red, green, yellow, or brown.
- They are filter feeders and eat tiny zooplankton.



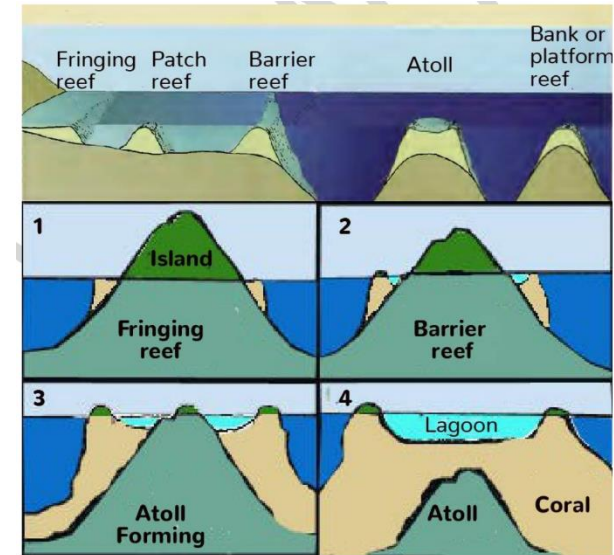
Types of Corals in News



- **Azooxanthellate Corals**
- Zoological Survey of India (ZSI) recorded four species of azooxanthellate corals from waters of Andaman and Nicobar (A&N) Islands.
- Azooxanthellate corals are a group of corals that do not contain zooxanthellae and derive nourishment not from the sun but from capturing different forms of plankton.
- They are deep-sea representatives, with the majority of species reporting from between 200 m to 1000 m. Their occurrences are also reported from shallow coastal waters.
- **Pillar coral (*Dendrogyra cylindrus*)**
- **Habitat:** Caribbean from the Yucatan Peninsula and Florida to Trinidad and Tobago.
- Moved from Vulnerable to Critically Endangered,

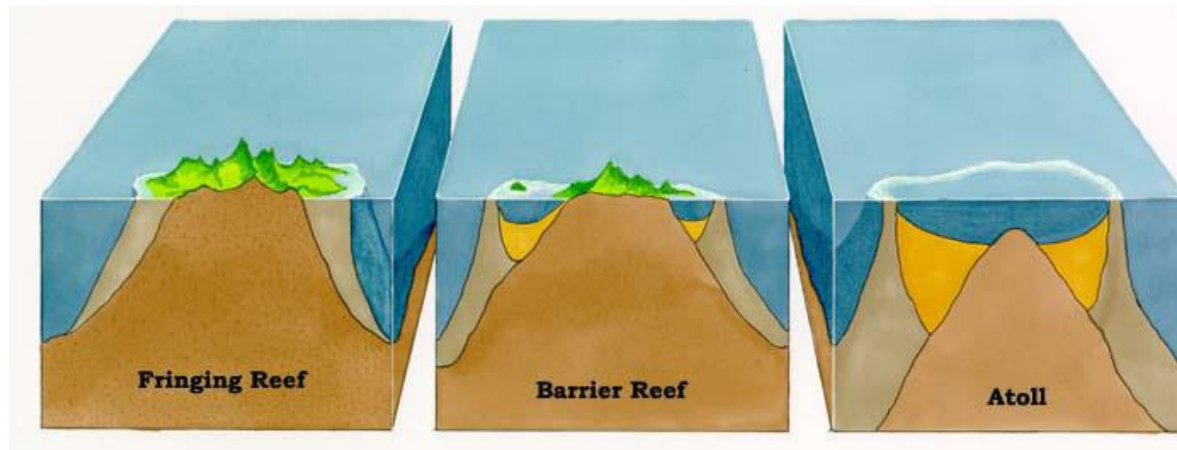
Coral Relief Features

- **Fringing Reefs (Shore Reefs)**
- Fringing reefs are reefs that grow directly from a shore. They are located very close to land, and often form a shallow lagoon between the beach and the main body of the reef. (most common)
- **Barrier Reefs**
- Barrier reefs are extensive linear reef complexes that parallel a shore and are separated from it by lagoon.
- Largest (in size, not distribution) of the three reefs
- Eg: Great Barrier Reef



Coral Relief Features

- **Atolls**
- An atoll is a roughly circular (annular) oceanic reef system surrounding a large (and often deep) central lagoon.
- A large 'number of atolls also occur in the Lakshadweep Islands.



Ideal Conditions for Coral Growth



- **Stable climatic conditions:** Highly susceptible to quick changes.
- **Perpetually warm waters:** Tropical waters (30°N to 30°S latitudes; temperature around 20°C) with narrow diurnal and annual temperature ranges. Coral reefs are absent on the west coast of tropical continents (Cold Ocean Currents).
- **Shallow water:** Ideal depths are 45 m to 55 m below the sea surface, where enough sunlight is available. Need good amount of sunlight to survive.
- **Clear salt water:** Clear salt water is suitable for growth, while freshwater & highly saline water are harmful.

Ideal Conditions for Coral Growth



- **Abundant Plankton:** Adequate supply of oxygen and microscopic marine food (plankton) is essential for growth. As the plankton is more abundant on the seaward side, corals proliferate on the seaward side.
- **Little or no pollution:** Corals are highly fragile and are vulnerable to climate change and pollution, and even a minute increase in marine pollution can be catastrophic.

Types of Corals

- **Hard corals:** primary reef-building corals, extract calcium carbonate from seawater to build hard, white coral exoskeletons. (Hermatypic Corals)
- **Soft Corals:** flexible because they lack solid skeleton (Ahermatypic Corals)

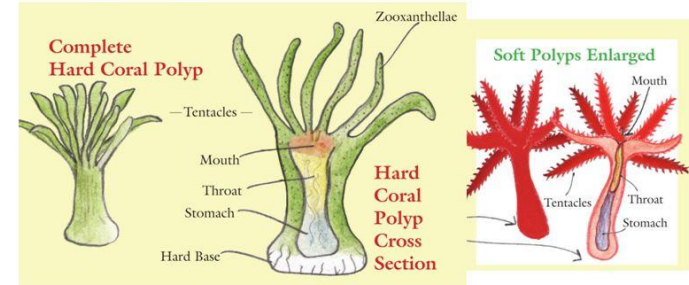
Hard corals vs. Soft corals

Hard corals

- Reef builders.
- Rigid skeleton made of calcium carbonate (CaCO_3)
- Secrete calcium carbonate

Soft corals

- No skeleton
- No calcium carbonate secretions



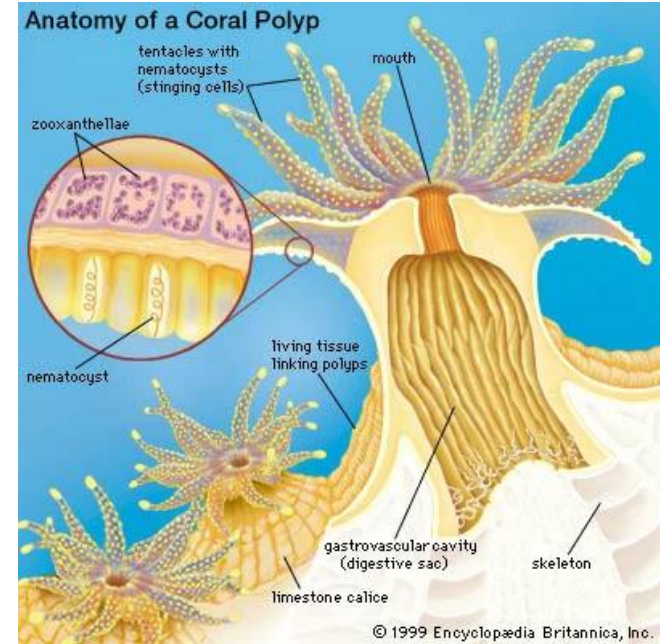
Hermatypic Corals (Corals and Zooxanthellae)



- Tropical Rainforests of the Oceans: Many invertebrates, vertebrates, and plants live in close association with corals, with tight resource coupling and recycling, allowing coral reefs to have extremely high productivity and biodiversity.
- Scleractinian corals (stony, hard corals) build skeletons of calcium carbonate sequestered from the water. They receive their nutrient and energy resources in two ways.
 - a. They use the traditional cnidarian strategy of capturing tiny planktonic organisms with their tentacles.
 - b. Having a symbiotic relationship with a single-cell photosynthetic alga known as zooxanthellae.

Corals and Zooxanthellae

- Zooxanthellae live symbiotically within the coral polyp tissues.
- The symbiotic relationship allows the slow-growing corals to compete with the faster-growing multicellular algae. The corals feed by day through photosynthesis (by zooxanthellae) and night through predation.
- The tissues of corals are inherently clear. The corals receive their colouration from the zooxanthellae living within their tissues.



Ahermatypic Polyyps



- Corals which live w/o symbiotic relations & Zooxanthellae.
- **They DO NOT BUILD REEFS.**
- It is not found in shallow waters but in ocean bottom floor. (Deep Ocean)
- It collects food directly from flowing water.
- It is found in cold, deep oceans. - Antarctica, Canada, Norway, US, UK, Japan, NZ.
- Fire Corals, Organpipe Corals, Black Corals (India)

Coral Bleaching



- Coral reef bleaching is a typical stress response of corals to various disturbances. Bleaching occurs when the densities of zooxanthellae decline and/or the concentration of photosynthetic pigments within the zooxanthellae fall (no longer helpful for the corals, and the corals will bleach them).
- During bleaching, they commonly lose 60-90% of their zooxanthellae, and each zooxanthellae may lose 50-80% of its photosynthetic pigments.
- **If the stress-causing bleaching is not too severe and decreases in time:** the affected corals usually regain their symbiotic algae within several weeks or a few months.
- **If zooxanthellae loss is prolonged:** zooxanthellae populations do not recover, the coral host eventually dies

Ecological Causes of Coral Bleaching



Temperature Changes

- Coral species (similar to most marine species) live within a relatively narrow temperature range.
- Coral bleaching events have occurred mainly during the protracted warming periods (summer seasons) and also during low wind velocity, clear skies, calm seas and low turbidity.
- These conditions favour localised heating and high ultraviolet (UV) radiation — which readily penetrates clear sea waters. (The corals **contain UV-absorbing compounds**, but rising temperatures mean a reduction in their concentration.)

Ecological Causes of Coral Bleaching



Ocean Acidification

- Acidification reduces corals' calcifying ability.

Subaerial exposure

- It can be caused extreme low tides, ENSO-related sea level drops, or tectonic uplift.
- Results in consequent exposure to high or low temperatures and increased solar radiation.

Inorganic Nutrients

- Rather than causing coral reef bleaching, an increase in ambient elemental nutrient concentrations (e.g., ammonia and nitrate) increases zooxanthellae densities 2-3 times.
- Although eutrophication is not directly involved in zooxanthellae loss, it could cause secondary adverse effects such as **lowering corals' resistance to diseases**.

Ecological Causes of Coral Bleaching



Xenobiotics

- Xenobiotics are chemical substances that are foreign to animal life. When corals are exposed to **high concentrations of chemical contaminants** like copper, herbicides and oil, coral bleaching happens.

Epizootics

- Most coral diseases cause patchy or whole colony death due to sloughing (shedding) of soft tissues, resulting in a white skeleton (not to be confused with bleached corals).

Great Barrier Reef



- The Great Barrier Reef, the world's largest reef system, is located in the Coral Sea, off Queensland, Australia.
- It is the world's most prominent single structure made by living organisms and can be seen from outer space. The reef is a habitat of the dugong (sea cow) and green turtle. It was selected as a World Heritage Site in 1981.
- According to IUCN, Australia's Great Barrier Reef is in a “critical state and deteriorating” as climate change warms up the waters in which it lies. Its IUCN conservation status has been downgraded from “significant concern” to “critical” because of frequent mass bleaching events (four in the past six years).
- An UN-backed mission recommended that the Great Barrier Reef be added to the list of endangered World Heritage sites (not yet).

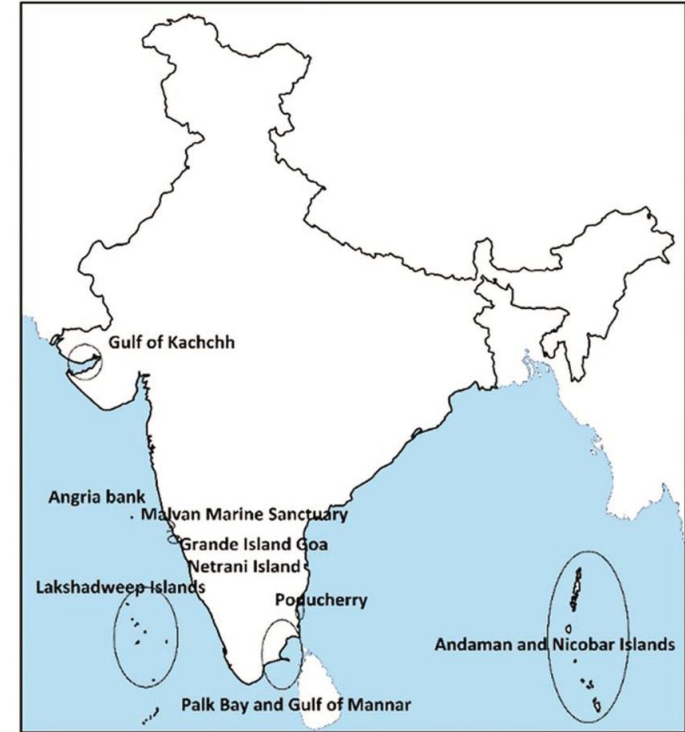
Great Barrier Reef

- A joint report by the IUCN and UNESCO's World Heritage Centre (WHC) has recommended that the GBR "be inscribed on the List of World Heritage in Danger".
- Inscribing a site on the Danger List allows the WHC to allocate immediate assistance from the World Heritage Fund, while simultaneously gathering international support and attention to the site.

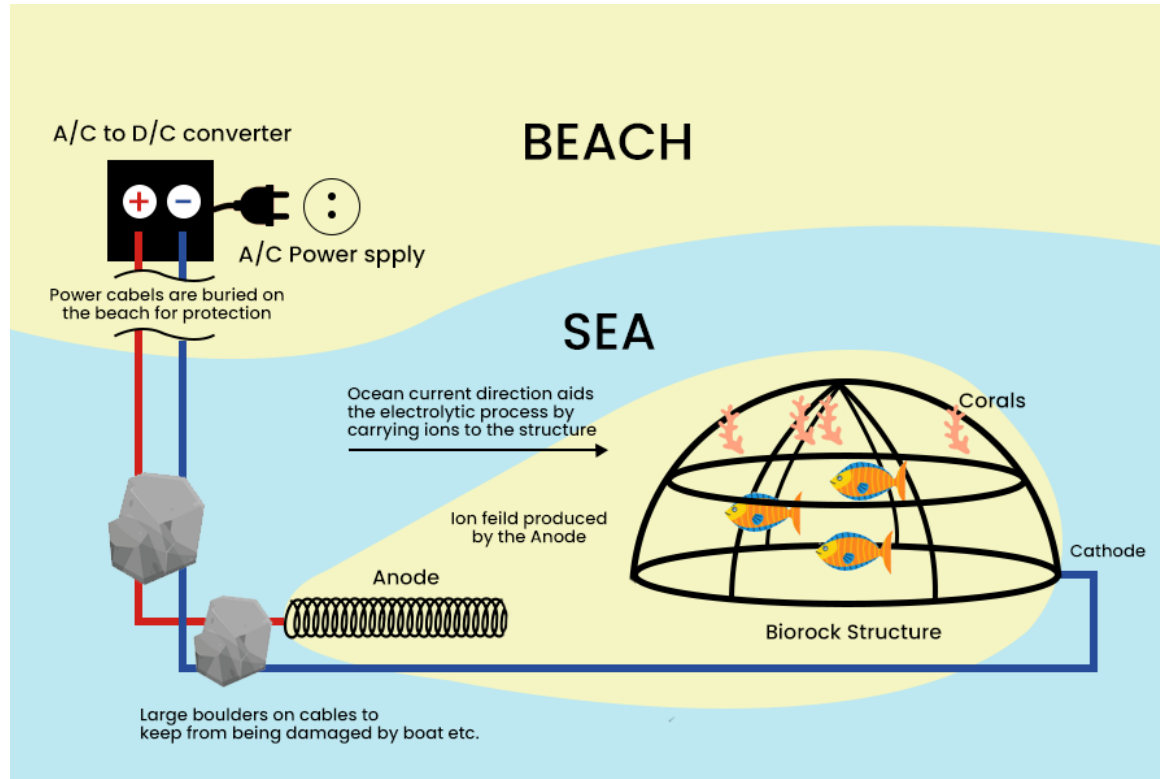


Coral Reefs In India

- The three major types of coral reefs found in India are fringing, barrier and atolls.
- Coral reefs in India are found in a lot of areas including the Gulf of Kutch, Gulf of Mannar, Palk Bay, Andaman & Nicobar and Lakshadweep Islands. The Gulf of Kutch in the northwest has some of the most northerly reefs in the world.
- Patches of coral reefs are also found in Ratnagiri, Malvan and Redi, south Bombay and at the Gaveshani Bank located in the west of Mangalore.



Biorock Technology for Coral Restoration



Biorock Technology for Coral Restoration



- The Zoological Survey of India attempted to restore coral reefs using biorock in the Gulf of Kachchh. If successful, biorock technology could potentially help to restore the degraded coral reefs.
- Biorock is the substance formed by mineral accretion or electro-accumulation of minerals on steel structures that are placed on the seabed and are connected to a power source like floating solar panels.

Biorock Technology for Coral Restoration



- The technology works by passing a small amount of electrical current through electrodes in the water. When electric current flows between the anode (+ve) and cathode (-ve) placed on the sea floor, calcium ions combine with carbonate ions and adhere to the cathode as calcium carbonate (CaCO_3). This electroaccumulated calcium carbonate is termed biorock.
- Fragments of broken corals are tied to the biorock structure, where the coral larvae adhere to the CaCO_3 and grow faster as they need not spend their energy building their CaCO_3 skeletons.

Cryomesh Technology

- Cryomesh Technology is to freeze and store coral larvae.
- Cryomesh method will help in building reefs threatened by climate change.
- Mesh technology will help to store coral larvae at -196°C (-320.8°F).



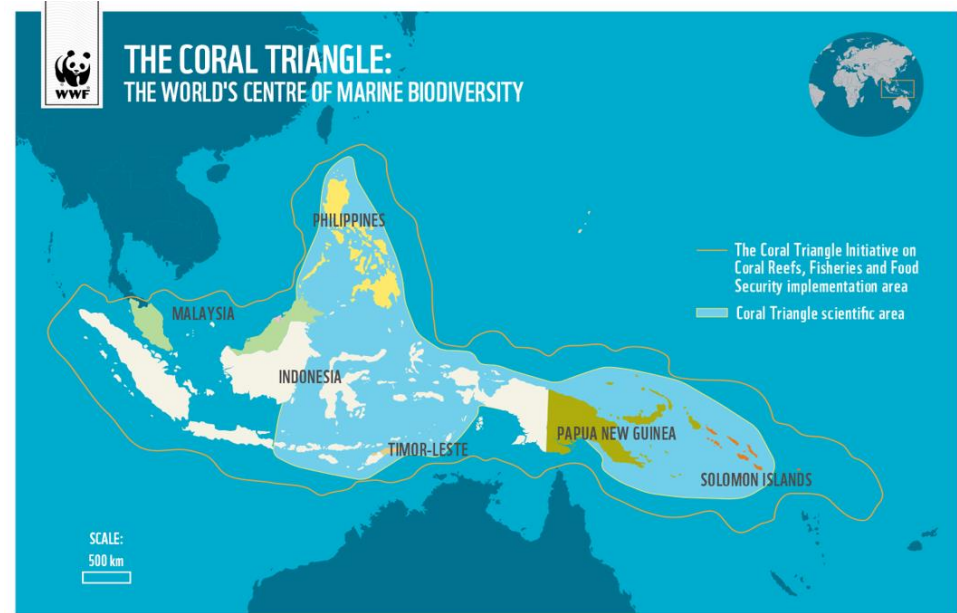
- **Coastal Zone Regulation**
- **Integrated Coastal and Marine Area Management (ICMAM)**
- **Coral reef areas:** On the recommendations of the National Committee on Mangroves and Coral Reefs following coral reef areas in the country have been identified for intensive conservation and management since 1987:
 1. Andaman & Nicobar Islands
 2. Lakshadweep Islands
 3. Gulf of Kutch (Gujarat)
 4. Gulf of Mannar (Tamil Nadu)

- The **Coral Bleaching Alert System**(INCOIS since 2011): uses the satellite derived Sea Surface Temperature (SST) in order to assess the thermal stress
- accumulated in the coral environs.
- National Coral Reef Research Centre has been established at Port Blair
- Blue flag Certification

- United Nations Environment Programme (UNEP) has included coral reef conservation and restoration as an **ecosystem based adaptation measure (EBA) for coastal protection.**
- **International Coral Reefs Initiative (ICRI)**
- Informal partnership between Nations and organizations India is a member of ICRI.
- The ICRI declared 2018 as the third International Year of the Reef (IYOR).
- **Global Coral Reef monitoring network** under ICRI: works to provide scientific information and communication on the status of coral reef ecosystems
- **Abu Dhabi Marine Restoration Initiative** (by The Environment Agency – Abu Dhabi)
- Aim: To restore coral, mangrove and seagrass in Abu Dhabi, creating a refuge for the dugong, a fast-disappearing aquatic mammal.

- **International Coral Reef Action Network (ICRAN)**
- It is a strategic alliance of private and public organizations that acts worldwide to
- address the management of coral reef ecosystems and the needs of the communities that depend upon them. shares and promotes traditional knowledge, current research, and best practices in order to strengthen reef management.
- **Coral Triangle Initiative**
- The six governments of the Coral Triangle – Indonesia, Malaysia, Papua New Guinea, the Philippines, Solomon Islands, and Timor Leste have established partnership to conserve coral reefs and the multitude of species and fisheries they support.

- **Coral Triangle**
- A marine area located in the western Pacific Ocean. It includes the waters of Indonesia, Malaysia, the Philippines, Papua New Guinea, Timor Leste and Solomon Islands.
- 200 varieties of reef fishes and supports livelihood of 120 million people.



Blue Flag Certification



- This Certification is accorded by an international agency “Foundation for Environment Education, Denmark” based on 33 stringent criteria in four major heads i.e
 - Environmental Education and Information,
 - Bathing Water Quality,
 - Environment Management and Conservation and
 - Safety and Services on the beaches.
- It started in France in 1985 and has been implemented in Europe since 1987, and in areas outside Europe since 2001 when South Africa joined.
- Spain tops the list with 566 such beaches; Greece and France follow with 515 and 395, respectively.

Blue Flag Certification

BLUE FLAG BEACHES OF INDIA



2022

Q. Consider the following statements :

1. High clouds primarily reflect solar radiation and cool the surface of the Earth.
2. Low clouds have a high absorption of infrared radiation emanating from the Earth's surface and thus cause warming effect.

Which of the statements given above is/are correct ?

- A. Only 1
- B. Only 2
- C. Both 1 & 2
- D. Neither 1 nor 2

2022

Answer- D

Whether a given cloud will heat or cool the surface depends on several factors, including the cloud's altitude, its size, and the make-up of the particles that form the cloud.

Statement 1 is incorrect: Low, thick clouds primarily reflect solar radiation and cool the surface of the Earth.

Statement 2 is incorrect: High, thin clouds primarily transmit incoming solar radiation; at the same time, they trap some of the outgoing infrared radiation emitted by the Earth and radiate it back downward, thereby warming the surface of the Earth.

2022

Among the following crops, which one is the most important anthropogenic source of both methane and nitrous oxide?

- [A] Cotton
- [B] Rice
- [C] Sugarcane
- [D] Wheat

2022

Answer- B

Methane and nitrous oxide are important greenhouse gases. They contribute to global warming. Important anthropogenic sources of biogenic methane are wet rice fields, cattle, animal waste, landfills and biomass burning.

Flooded rice cultivation has been identified as one of the leading global agricultural sources of anthropogenic methane (CH_4) emissions. Furthermore, it has been estimated that global rice production is responsible for 11% of total anthropogenic CH_4 emissions.

Paddies are a potential source of anthropogenic nitrous oxide (N_2O) emission as well. In paddies, both the soil and the rice plants emit N_2O into the atmosphere. The rice plant in the paddy is considered to act as a channel between the soil and the atmosphere for N_2O emission.

2020

Which one of the following statements best describes the term ‘Social Cost of Carbon’? It is a measure, in monetary value, of the

- a) long-term damage done by a ton of CO₂ emissions in a given year.
- b) requirement of fossil fuels for a country to provide goods and services to its citizens, based on the burning of those fuels.
- c) efforts put in by a climate refugee to adapt to live in a new place.
- d) contribution of an individual person to the carbon footprint on the planet Earth.

Answer: A

- The economic losses suffered due to the emission of one ton of carbon dioxide into the atmosphere is termed the social cost of carbon. It is expressed as the dollar value of the losses suffered.

2019

Why are dewdrops not formed on a cloudy night?

- [A] Clouds absorb the radiation released from the Earth's surface.
- [B] Clouds reflect back the Earth's radiation.
- [C] The Earth's surface would have low temperature on cloudy nights.
- [D] Clouds deflect the blowing wind to ground level.

2019

Answer- B

Because on a cloudy night, the clouds send the heat back to the ground so the ground never gets cold enough for the dew to be formed.

Dew drops are formed due to condensation of water vapours. Air around us contains water vapours which we call moisture or humidity. Hot air contains more moisture as compared to cold air.

During the night when the hot air comes into contact with some cold surface due to the reflection of earth radiation by clouds, water vapour present in it condenses on the cold surface in the form of droplets. These tiny drops of water are called dew drops.

2018

Which of the following statements best describes “carbon fertilization”?

- a) Increased plant growth due to increased concentration of carbon dioxide in the atmosphere
- b) Increased temperature of Earth due to increased concentration of carbon dioxide in the atmosphere
- c) Increased acidity of oceans as a result of increased concentration of carbon dioxide in the atmosphere
- d) Adaptation of all living beings on Earth to the climate change brought about by the increased concentration of carbon dioxide in the atmosphere

Answer: A

- The carbon dioxide fertilization or carbon fertilisation is not only the cause of plant growth but also contributes the greening effect.
- It is the phenomena that the increase of carbon dioxide in the atmosphere increases the rate of photosynthesis in plants.

2019

Consider the following:

1. Carbon monoxide
2. Methane
3. Ozone
4. Sulphur dioxide

Which of the above are released into the atmosphere due to the burning of crop/biomass residue?

- a) 1 and 2 only
- b) 2, 3 and 4 only
- c) 1 and 4 only
- d) 1, 2, 3 and 4

Answer:

- O₃ is not released directly. It is a secondary pollutant.
- Burning of biomass releases CO, CH₄ and SO₂.

2011

The increasing amount of carbon dioxide in the air is slowly raising the temperature of the atmosphere, because it absorbs

- a) the water vapour of the air and retains its heat.
- b) the ultraviolet part of the solar radiation.
- c) all the solar radiations.
- d) the infrared part of the solar radiation

Answer:

D

- Among GHGs, only water vapour can absorb both incoming (UV) and outgoing (infrared) radiation.

2011

Consider the following statements: Chlorofluorocarbons, known as ozone depleting substances, are used

1. in the production of plastic foams
2. in the production of tubeless tyres
3. in cleaning certain electronic components
4. as pressurising agents in aerosol cans

Which of the statements given above is/are correct?

- a. 1, 2 and 3 only
- b. 4 only
- c. 1, 3 and 4 only
- d. 1, 2, 3 and 4

2011

Answer: c)

Explanation:

- CFCs were used as refrigerants, pressurising agents (aerosol cans) & for cleaning electronic equipment.

2011

The formation of ozone hole in the Antarctic region has been a cause of concern. What is the reason for the formation of this hole?

- a) Presence of prominent tropospheric turbulence; and inflow of chlorofluorocarbons
- b) Presence of prominent polar front and stratospheric clouds; and inflow of chlorofluorocarbons.
- c) Absence of polar front and stratospheric clouds; and inflow of methane and chlorofluorocarbons.
- d) Increased temperature at polar region due to global warming

2011

Answer: B)

During the winter, temperatures inside the Antarctic polar vortex fall so low that water vapour and several other types of molecules in the stratosphere condense into extremely small icy particles. These icy particles, in turn, make up polar stratospheric clouds (PSCs). When the sun sets in the Antarctic around the end of March each year, its disappearance marks the beginning of a long, dark winter. Once the last rays of sunlight have faded away, temperatures on land and in the air fall very quickly.

In the stratosphere, high-altitude winds that create the polar vortex begin to blow around the continent. Isolated from warmer air outside the vortex, the air inside gets colder and colder. Eventually, it is cold enough for PSCs to form.

2011

The increasing amount of carbon dioxide in the air is slowly raising the temperature of the atmosphere, because it absorbs

- a) the water vapour of the air and retains its heat.
- b) the ultraviolet part of the solar radiation.
- c) all the solar radiations.
- d) the infrared part of the solar radiation

2011

Answer: d) the infrared part of the solar radiation (outgoing radiation).

Explanation:

- Among GHGs, only water vapour can absorb both incoming (UV) and outgoing (infrared) radiation.